

Definition Breaker

DB-1.0 - Child Access Standard and Inclusive Play System

A major child-accessibility upgrade for digital games, classroom learning, communication, physical toys, family play and later expert study of the complete UGTS substrate.

Notice

See, hear, feel or receive a clear summary.

Choose

Point, look, sign, speak, scan, press or ask a partner.

Control

Stop, wait, repair, change route and return.

Preview

Try a candidate world before it becomes real.

Check

Learn the first failed rule in plain language.

Remember

Replay what changed without a memory test.

WHERE? MATCH? READY? CHECKED? CHANGE REMEMBER

Evidence and access boundary. UGTS mechanics are grounded in the supplied substrate reports. Child-access, communication, learning, toy and co-design requirements are design-derived and informed by current accessibility guidance. This PDF targets tagged PDF/UA-1 output and has been technically preflighted, but independent PDF/UA, educational, clinical, toy-safety and regulatory certification are not claimed.

Edition

Executive decision: access is part of the rules

DB-1.0 treats accessibility as a first-class route through the same authoritative game world. It does not create a lower-status mode, hide an adult inside the controls, require a diagnosis, or make children earn access by completing an inaccessible tutorial.

The child directs access

The child may select, combine, pause or replace visual, auditory, tactile, motor, communication, cognitive, regulation and energy routes at any time.

The concept stays whole

Support, compatibility, guard, verified event, transition and lineage remain the learning goal. Access changes channel, pace or amount shown - not the idea.

Assistance can be valid play

AAC, switch scanning, eye gaze, co-play and partner-assisted selection are legitimate routes when the child retains choice, confirmation and repair.

Failure becomes information

A rejected action identifies the first unmet check and offers repair. It never labels the child as wrong, slow, inattentive or incapable.

Access survives interruption

A break, tracking loss, device change, fatigue shift or sensory change preserves focus, candidate state, replay and the next available choices.

Exclusion blocks release

An essential action with only one sensory or motor route, inaccessible help, forced touch, hidden partner substitution or unequal reward is a release-blocking defect.

Decision: GO

Adopt DB-1.0 as the accessibility baseline for Definition Breaker digital play, learning materials and physical prototypes. Every later implementation must provide evidence of child agency, route parity, repair and safe return.

The DB-1.0 access envelope

An action is not accessible merely because a child can activate a button. The complete route must let the child perceive, understand, act, confirm, repair and resume.

1 Perceive

receive the object, relation and state

2 Understand

know the purpose and likely result

3 Act

express an intent with a workable route

4 Confirm

verify that the system understood

5 Repair

cancel, undo or reinterpret safely

6 Resume

return after interruption without a memory test

Semantic parity

Different routes expose the same object identity, relationships, state and available actions.

Outcome parity

The same child intent reaches the same proposal, verification, commit, reward and story consequence.

Dignity parity

An accessible route does not use babyish presentation, public labels, pity, lower rewards or unnecessary adult control.

Effort proportionality

The access route should not demand far more repetitions, navigation, posture, attention or fatigue than the standard route.

Release rule

Every essential game, learning, communication and physical-play action must pass all six access-envelope stages through at least one route that works for the child in that session.

What is greatly strengthened

DB-0.6 established combined access, deafblind routes, AAC, switch and eye-gaze pathways, touch consent and child-defined metrics. DB-1.0 turns those principles into a complete operating standard.

Strengthening matrix

Area	Retained baseline	DB-1.0 strengthening
Access model	Many valid routes	The six-part access envelope plus semantic, outcome, dignity and effort parity tests.
First run	Accessible onboarding	A no-wrong-door discovery sequence with STOP and HELP available before calibration.
Communication	AAC and partner support	Core-plus-fringe vocabulary, micro-signal observation, interpretation repair and partner-fidelity testing.
Input	Switch, gaze and one-hand routes	One-, two- and step-scan profiles; dwell rest zones; device-loss continuity; non-drag and non-hold alternatives.
Nonvisual play	Screen reader and tactile routes	Queryable graph summaries, landmark grammar, object-state narration and tactile replay parity.
Learning	Child language for the substrate	Same-idea/three-language pages connecting child, facilitator and formal vocabulary.
Physical play	Accessible prototypes	Low-force mounting, tactile symbol grammar, object cues, household parity and product-development scaffolds.
Evidence	Release gates	Combined-access scenarios, time-to-stop, repair success, route-switch continuity and child-agency metrics.
Privacy	AccessProfile separation	No diagnosis inference, emotion scoring, accessibility paywall or child-data training by default.

Supersession rule

Every DB-0.6 requirement remains in force unless DB-1.0 gives the child more agency, more reliable communication, less forced effort, stronger privacy or a more equal route to the same concept.

How to use this edition

A child, family member, teacher, therapist, toy designer and game engineer may enter at different pages while sharing one vocabulary. Reading the whole document is not a prerequisite for using it.

Child playbook

Rights, choice, STOP, communication, switch, gaze, touch, listening, safe mistakes, return and replay.

Communication and agency

AAC, no-response policy, early communication, supported decision-making and partner fidelity.

Combined access

Functional access needs, including combined and fluctuating routes.

Learning the substrate

The complete architecture in child language without replacing the formal ideas.

Digital product standard

Semantic objects, input adapters, nonvisual queries, feedback, privacy and accessible replay.

Physical play

Toys, tabletop games, object cues, tactile schedules and accessible group play.

Evidence and release

Co-design, safeguarding, tests, metrics, linting, release gates and kill criteria.

Printables

Large, high-contrast prototype boards for communication, scanning, rules and replay.

Reading is not a test

A reader may listen, magnify, print, skip, use a screen reader, receive a tactile explanation, stop, or ask another person to relay the content. Eye contact, speech, fast response and independent hand use are never prerequisites.

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My access rights

This game must work with me. It must not test whether I can act, speak, read, move or stay calm like somebody else.

I choose my route

I may look, listen, touch, point, sign, speak, scan, press, wait or use a partner.

I can say no

STOP, NOT THAT, WAIT, QUIET, BREAK and DONE work immediately.

I can change my mind

A choice is not a trap. I may repair it before or after preview.

I get the real idea

My access route does not move me into a lesser puzzle, story, role or reward.

I get time

Thinking, looking away, moving slowly, repeating or not responding yet are allowed.

I can come back

The game tells me where I was. Returning is not a memory test.

My communication counts

A sound, movement, gaze, sign, object, switch or partner-confirmed signal may be my answer.

I can ask what is stored

My access choices are private and must not be used to rank, diagnose or advertise to me.

My rule

Nothing becomes real until I can understand the preview, the action is checked, and I still have a way to stop or repair it.

STOP and HELP work first

Before a title screen, story, calibration, login or tutorial asks anything of me, I need two reliable actions: STOP and HELP.

STOP

Pause sound, motion, scanning and world changes. Keep my place. Do not ask why.

HELP

Offer a short explanation, show another route, or call a trusted person without taking control.

WAIT

Give me more time. Do not repeat louder or faster unless I ask.

QUIET

Reduce music, effects, speech, vibration, animation and visual clutter immediately.

NOT THAT

Cancel the current interpretation. Return to the last clear choice.

DONE

End safely, save the checkpoint, and show how to return later.

STOP is reachable by: keyboard, controller, touch, switch scan, eye gaze, screen reader action, AAC or partner command, and a physical stop card.

No penalty

Stopping, changing route, using help, slowing down or playing with support never lowers score, story access, badge status or reward.

A no-wrong-door first run

The first two minutes help the system learn what works without asking me to prove a disability or finish a calibration test.

1 Welcome

one calm sentence and one object

2 Stop

show and test STOP

3 Choose

offer look, listen, touch, press, scan or partner

4 Try

one reversible cause-and-effect action

5 Confirm

ask whether that route felt right

6 Save

remember preferences locally and privately

Every route reaches settings

No inaccessible splash screen, captcha, hold gesture, tiny icon or audio-only prompt stands between me and access controls.

Calibration is optional

Gaze, switch and sound calibration may be skipped, repeated, slowed or completed with support and my confirmation.

Suggestions are reversible

The system may offer a route after repeated misses or fatigue. It does not silently change difficulty or label me.

The first success matters

The opening action changes something I can perceive through my route and can undo immediately.

First-minute success

I can stop, make one meaningful choice, receive the result and repair it without losing progress.

Choose what helps now

Pick one, several or none. You may change the answer later.

LOOK

Large text, simple picture, strong border and clear focus.

LISTEN

Spoken choices, scene summary and change replay.

TOUCH

Real objects, raised lines, textures and a stable board.

POINT OR GAZE

Finger, eye, head, pointer, gesture or card.

PRESS OR SCAN

One key, one switch, two switches, dwell or partner scan.

SIGN OR AAC

Sign, symbol, speech device, communication book or custom signal.

PARTNER

A person names or moves only the choice I confirm.

QUIET OR SLOW

Less motion, less sound, fewer choices and more time.

Today I choose:

I can answer in many ways

The game records my chosen action, not a judgement about the body movement used to express it.

LOOK

Steady gaze, shift of gaze or look-away choice.

TOUCH

Tap, hold, move toward, move away or release.

MOVE

Head, hand, foot, body, breath or facial signal.

PRESS

Key, switch, touch zone or dwell.

SIGN

Formal sign, gesture, tactile sign or object cue.

SAY

Speech, sound, vocalization or speech device.

PARTNER

Someone notices, waits, confirms and then relays.

NOT YET

No response can mean more time or another route.

Every route counts

The same answer produces the same candidate proposal, preview, check and reward.

Stop, change and repair

These actions stay available before, during and after every preview.

STOP

End the current action.

NOT THAT

That was not my choice.

WAIT

Give me more time.

AGAIN

Show, say or present it again.

CHANGE ROUTE

Use another way.

I CHANGED MY MIND

Undo and let me choose again.

HELP

Support me without choosing for me.

DONE

Save and finish.

Repair is normal

A wrong interpretation is a system-and-partner repair task, not evidence that I cannot communicate.

First, then, next, break, finished

I can see what is happening now and what will happen after it.

1 First

find one thing

2 Then

try one change

3 Next

preview or check

4 Break

pause and keep my place

5 Finished

save and close

Show less

Fewer objects, one question, less movement, quiet cues or one scan group.

Show more

Formal words, full dependency graph, extra examples or a detailed replay.

Keep my place

Mark the current object, current step and last checked change.

Tell me about changes

Warn before a new room, person, sound, movement or rule.

Important

Showing less information is not the same as teaching a lesser concept. Hidden information remains stable and can be revealed when useful.

Know before the room starts

The room tells me what it may do. I choose changes before I enter.

Sensory forecast choices

Channel	This room may use	My choice
Sound	music, voice, effects, silence changes	off / quiet / captions / visual or tactile cue
Light	bright areas, contrast changes, glow	soft / high contrast / no glow / nonvisual route
Motion	camera movement, animation, auto-scroll	still / reduced / slow / follow on request
Touch	vibration, textured pieces, partner touch	off / gentle / tactile object / ask first
Time	scanning, countdown, repeated prompt	untimed / slow / pause / practice
People	solo, partner, group, turn-taking	quiet role / co-play / solo / observe first

I want: quiet / slow / still / touch / listen / other

My body and energy may change

The game believes me when I say that today is different from yesterday or ten minutes ago.

READY

I can start.

SLOW

Use fewer steps and more time.

QUIET

Reduce sound, motion and choices.

SUPPORTED

Use switch, gaze, partner or positioning help.

BREAK

Save my place now.

CHANGE

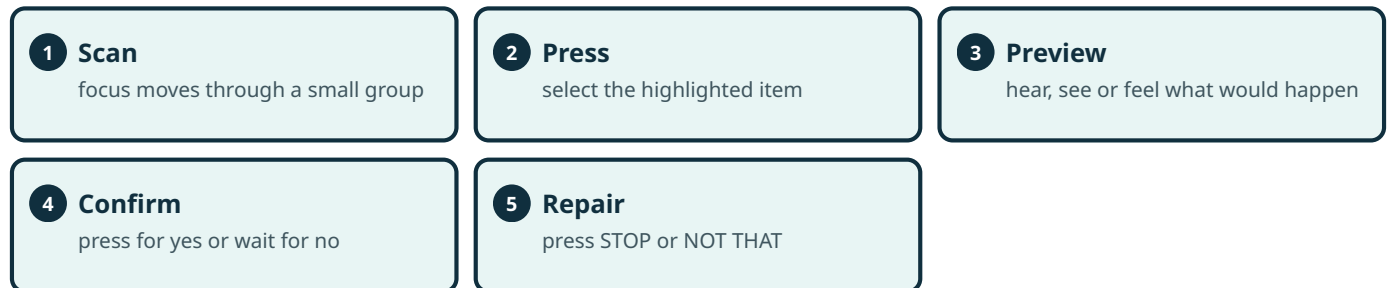
Move to another access route without restarting.

The game may notice; it may not diagnose

Long pauses, repeated accidental activation or tracking loss may trigger an offer such as "Would another route help?" They do not change score, infer a condition or report private health information.

Play with one button

One switch is enough to explore, choose, confirm, stop and repair when the interface scans predictably.



- No choice disappears while I am thinking.
- Scan speed, order, grouping, cue channel and confirmation are adjustable.
- STOP and HELP appear in every scan loop.
- Accidental presses are reversible and do not reset the room.
- Holding a switch is never required.

Rest is allowed

A pause in activation does not mean I have finished. The scan can wait on a neutral rest item until I am ready.

Play with two buttons or step scan

A second switch may separate moving from choosing, but I can always return to one-switch mode.

NEXT + CHOOSE

One switch moves focus; the other selects.

LEFT + RIGHT

Move through a short row in both directions; dwell or another action confirms.

YES + NO

Use two reliable signals after a preview.

STEP SCAN

I advance only when ready; the system never races.

GROUP + ITEM

Choose a group, then an item; back and stop remain available.

PARTNER STEP

The partner advances when asked; I confirm the item.

No route hierarchy

Two switches are not “better” than one. We choose the route with the lowest effort and clearest control for this session.

Play with eye gaze or head pointing

The system does not treat every glance as a command.

Large stable targets

Targets remain in the same place, have wide spacing and do not move while I choose.

Rest zone

A neutral area lets me look without selecting.

Dwell preview

A visible, audible or tactile progress cue shows when a choice is forming.

Confirm or cancel

Important changes use a second look, switch, yes/no choice or partner confirmation.

Tracking loss

Pause safely and keep focus. Do not guess a choice.

Calibration repair

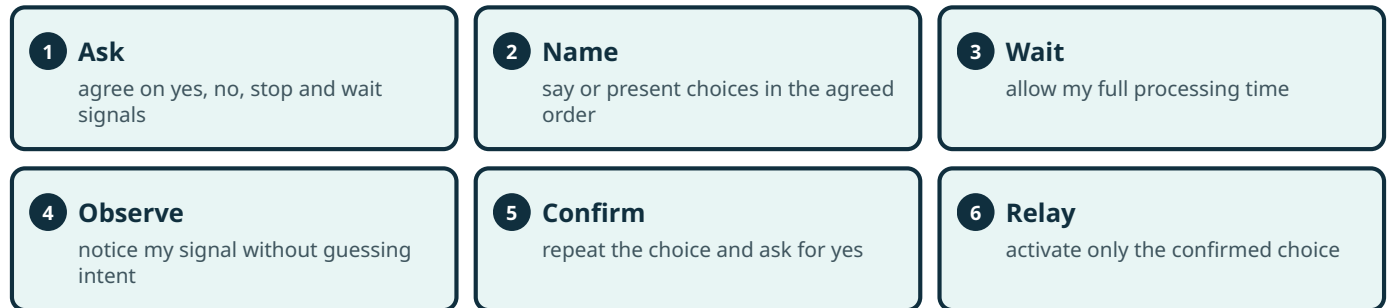
Offer quick recalibration, target enlargement, head pointing or partner scan without restarting.

Midas-touch rule

Looking to understand is not the same as choosing. Reading, exploring and resting never trigger a commit by themselves.

A partner names; I decide

The partner is an access adapter, not the hidden player.



- Use the same wording and order each round unless I ask for change.
- Do not skip a choice because an adult thinks it is unhelpful.
- Record uncertainty as uncertainty; do not convert it into yes.
- I may reject the partner's interpretation at any time.
- Another partner can reproduce the protocol from my Access Passport.

Learn through touch

Touch is offered, never forced. I may use real objects, raised symbols, textures, braille, vibration or a partner description.

Home corner

One stable orientation marker anchors every board.

One object, one role

A rule, gate, route and replay marker have distinct shapes or textures.

Explore before asking

I get time to examine pieces without turning exploration into a quiz.

Hand-under-hand

When invited, an adult supports beneath or beside my hand rather than moving it from above.

Tactile preview

Candidate pieces go on a separate try mat before moving the real world.

Tactile replay

A bead or token chain records each checked change in order.

Touch consent

Ask before touch, describe what will happen, wait for my signal, stop immediately on withdrawal or refusal, and offer a non-touch route.

Learn through listening

Sound gives meaning only when it has a stable, optional legend and a non-audio equivalent.

Scene summary

Name the room, important objects, current focus and available actions.

Object details

On request, describe role, state, links, location and first unmet check.

Stable earcons

A short sound may mark support, match, ready, checked, change or remember.

Speech controls

Rate, voice, detail, repetition, interruption and pronunciation are adjustable.

Spatial audio

Use relative directions only with a recenter action and text or tactile equivalent.

Quiet route

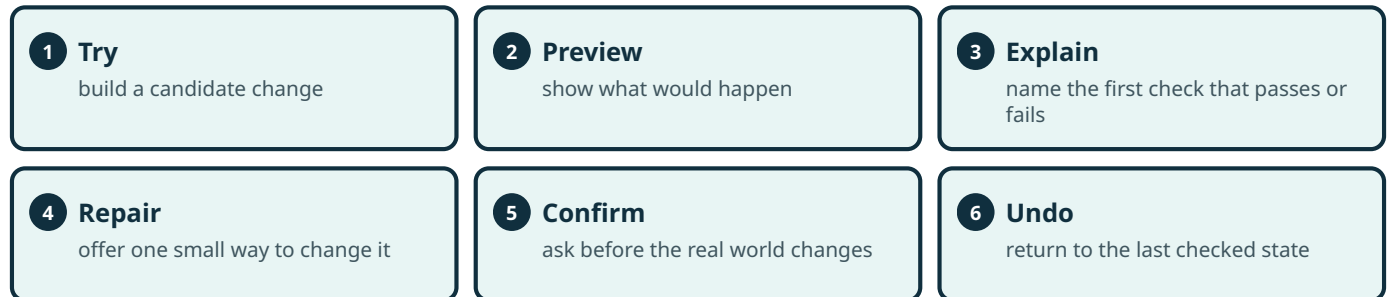
Every sound can be reduced, captioned or replaced without losing information.

No sound-only puzzle

Pitch, rhythm, stereo position or speech alone never carry an essential rule unless sound is the specific learning goal and an equivalent explanation is provided.

A mistake stays safe

The world does not punish me for exploring a rule.



No shame language

Use “not ready yet,” “these pieces do not match,” or “this route is blocked,” not “wrong,” “bad,” or “you failed.”

Unlimited rehearsal

Practice, replay and hints do not lower status or reward.

Visible uncertainty

When the system is unsure, it says so and offers another check.

Adult restraint

A supporter may explain or move a proposed piece, but I still confirm the choice.

Come back without a memory test

The game rebuilds context so I do not have to remember every prior step.

Where I was

Name the room, goal and current focus.

What changed

Replay only the important committed events in my chosen channel.

What I was trying

Restore the candidate proposal separately from the real world.

What can happen next

Offer a small set: continue, undo, review, change route or finish.

What helped

Restore access preferences locally, with a quick way to change them.

What stays private

Do not announce diagnosis, access profile or communication method to other players.

Return promise

I can leave at any time, come back on another device or route, and continue from a meaningful checkpoint without penalty.

Tell me what changed

Replay can be words, touch, sound, symbols, animation, a partner explanation or a mixture.

Replay vocabulary

Replay question	Example child answer	Formal record
Where?	The bridge rule reached the left platform.	Support result
Match?	The triangle plug and triangle socket can connect.	Compatibility result
Ready?	The gate crossed its safe line.	Guard status and margin
Checked?	All required checks passed.	Verified proposal
Change?	The bridge moved before it turned.	Atomic patch and operation order
Remember?	This was the third checked change.	Sequence, checkpoint and lineage

Detail control

I may ask for one sentence, a step list, the full dependency trace or the first divergence. More detail is available; it is not forced.

My Access Passport

This page helps another person or device offer access in a familiar way. It is not a diagnosis, test score or permanent label.

Access Passport fields

Area	My preference
How I say YES / NO / STOP	
How long I usually need to answer	
Ways I like to receive information	
Ways I like to choose or move	
Touch and positioning preferences	
Sound, light, motion and vibration choices	
What helps when I am tired or overloaded	
What a partner must confirm with me	
What must stay private	

I own this information

Review it with me. Let me change, remove or refuse entries. Store only what is necessary for access.

Communication autonomy is game authority

A child's communication method is not a side setting. It is the route through which proposals, confirmations, refusals, questions and repairs enter the authoritative system.

Assume competence

Begin from the possibility that behavior has meaning. Observe, offer choices and confirm rather than withholding access until a child proves symbolic language.

Model, do not demand imitation

Adults may model symbols, signs, objects or AAC without requiring the child to copy them before continuing.

Keep core words available

STOP, MORE, AGAIN, DIFFERENT, HELP, WAIT, YES, NO, NOT SURE, DONE and CHANGE ROUTE remain reachable in every room.

Preserve the child's voice

A partner may relay but may not rewrite, soften, embellish or replace the child's choice.

Honor unconventional signals

Gaze, breath, muscle tone, approach, withdrawal, vocalization and repeated body patterns may become agreed signals after careful confirmation.

Repair openly

When interpretation fails, record "unclear" and renegotiate the signal instead of blaming the child.

Core principle

Communication access is available before instructions, consent, assessment and scoring. It cannot be a reward unlocked after compliance.

Core words plus puzzle words

Definition Breaker needs a stable communication layer and room-specific vocabulary. Existing AAC systems remain primary; the game supplies compatible named actions and low-tech backups.

- Keep symbol locations stable across rooms.
- Pair symbols with words and speech when useful; do not require literacy.
- Allow the child's existing AAC system to send named actions.
- Do not lock communication vocabulary behind gameplay menus.
- Support multilingual vocabulary and the child's preferred language.

AAC core and fringe vocabulary

Always-available core	Puzzle fringe	System response
STOP / WAIT / HELP	bridge / gate / sheet / seed	Immediate control or object reference
YES / NO / NOT SURE	support / match / ready	Confirmation with uncertainty allowed
MORE / LESS / AGAIN	detail / clue / replay	Adjust amount or repeat without penalty
DIFFERENT / CHANGE	route / order / rule	Offer alternatives or edit candidate
MINE / YOURS / TOGETHER	owner / role / turn	Co-play and ownership choice
DONE / BREAK / COME BACK	save / checkpoint	Safe exit and return

Yes, no and not sure are all real answers

A binary interface must not force an uncertain child into yes or no.

YES

Commit or continue only after the child's agreed yes signal.

NO

Cancel without pressure, disappointment animation or repeated persuasion.

NOT SURE

Keep the proposal in preview, provide more information or offer another route.

CHANGED MIND

Undo the last choice and return to the decision point.

NO RESPONSE YET

Wait, reduce options, check comfort or pause; do not infer agreement.

PARTNER UNSURE

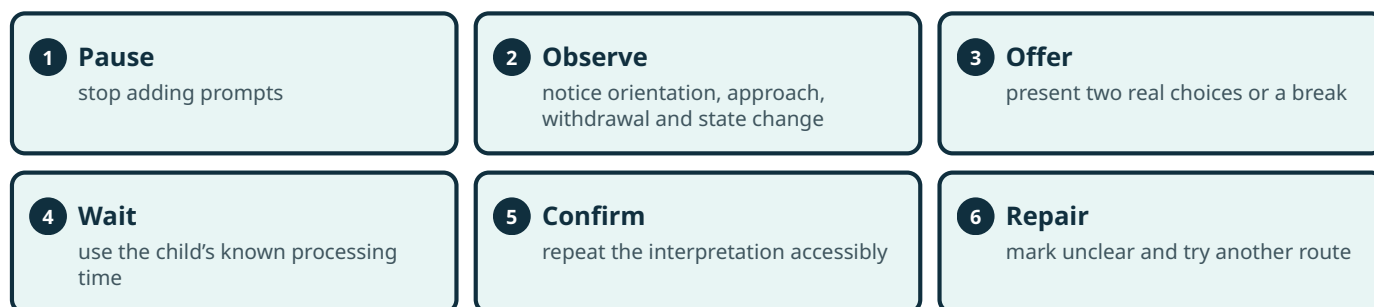
The partner states uncertainty and seeks another confirmation method.

Consent rule

Silence, stillness, compliance, lack of resistance or an adult's confidence is not the child's yes.

No response is not failure

A child may need more processing time, a different cue, a different body position, a communication partner, a break or a different route.



- Do not repeat the same demand faster or louder.
- Do not remove the child from the activity for slow response.
- Do not score “no response” as incorrect.
- Do not convert a supporter’s prediction into the child’s answer.
- Do not use compliance as the only evidence of understanding.

The concept can begin before symbolic language

The first learning goal may be shared attention, anticipation, preference, cause-and-effect or turn-taking while the same substrate structure remains available to grow into.

Real effect

One reliable signal changes one perceivable part of the candidate world.

Real choice

Both offered options are possible and respected; one is not a disguised test.

Anticipation

A stable object cue, tactile schedule or sound prepares the child for what comes next.

Shared attention

The adult follows the child's focus and builds a turn around it.

Preference history

Repeated choices inform future offers but never become an irreversible label.

Growth path

Object -> action -> symbol -> rule name -> dependency -> formal trace, without requiring every child to use every step.

No passive-access substitute

Sensory stimulation without choice, consequence or repair is not equivalent to playing the rule game.

Help the child decide; do not decide for the child

Support improves access to information, options and communication while keeping the choice with the child.

Supported decision-making boundaries

Supporter may	Supporter must not
Explain one option in a preferred channel.	Remove options because they seem difficult or inconvenient.
Move or hold a piece at the child's direction.	Treat physical assistance as permission to choose.
Offer extra time, examples or a practice round.	Demand a faster answer to keep the group moving.
Repeat the child's signal and ask for confirmation.	Claim certainty when the signal is unclear.
Record access preferences with permission.	Publish, diagnose or rank the child from those preferences.
Pause when the child shows refusal or overload.	Use rewards, guilt or peer pressure to obtain compliance.

Measure agency, not independence

A supported choice can be fully child-directed. Independence is one possible route, not the definition of success.

The partner is an ethical access adapter

A person can unintentionally change order, timing, wording and outcomes. Partner assistance must therefore be inspectable and testable like any other input adapter.

Stable protocol

Document choice order, cue words, yes/no/stop signals, wait time and confirmation method.

Neutral presentation

Use similar tone, duration and visibility for each genuine option.

No silent filtering

Present all relevant choices, including stop and different route.

Observable confirmation

The committed action must be traceable to an agreed child signal.

Repair log

Record interpretation changes without storing unnecessary private body data.

Partner rotation test

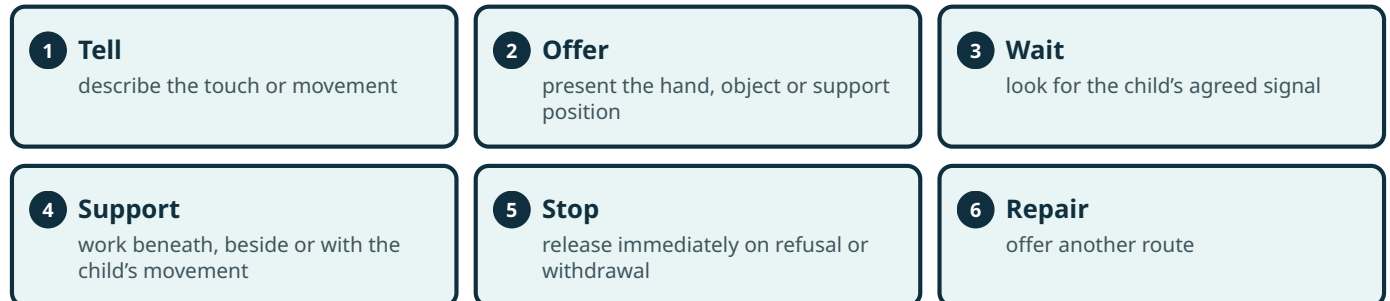
A second trained partner should reproduce the process without changing the child's decisions.

Release blocker

If an adult can make the game appear accessible while the child's choices do not affect the authoritative proposal, the route fails.

No surprise touch and no forced hand movement

Tactile learning can be powerful, but touch is personal and remains under the child's control.



- Never take the child's hand and complete the puzzle for them.
- Never restrain withdrawal to finish a turn.
- Use object cues before moving equipment or changing body position.
- Document individual touch preferences without treating them as permanent.
- Collaborate with trained deafblind and occupational specialists for child-specific methods.

Language access is disability access

A child should not have to learn the designer's language before learning the rule.

Preferred language first

Instructions, core communication and safety actions should be available in the child's strongest language.

Symbols are not universal

Test icons and tactile marks with the child and community; keep the accompanying word or explanation.

Sign languages are languages

Provide room for signed video or live interpretation; do not treat sign as word-for-word gesture coding.

AAC supports multiple languages

Do not remove home-language vocabulary to make a board simpler.

Text expansion

Layouts tolerate longer words, right-to-left scripts and different line breaks.

Name and identity

Use the child's chosen name, pronouns and culturally respectful examples without requiring public disclosure.

Learning parity

A translation may adapt examples and sentence order, but it preserves the same support, compatibility, guard and transition relationships.

Accessible does not mean childish

Children with disabilities deserve humor, challenge, style, privacy and stories that fit their interests - not a permanently infantile interface.

Interest-led themes

Offer science, fantasy, engineering, art, animals, vehicles, music, space and other themes without ability labels.

No baby mode

Simpler language, fewer objects or slower pacing must not force babyish art or exaggerated praise.

Private access tools

Other players do not need to see scan speed, AAC route, diagnosis or helper status.

Respect regulation

Harmless stimming, gaze aversion and movement are not failure.

Avoid hero or tragedy narratives

Disability is not a plot reward, punishment or inspiration device.

Credit the child

Badges and histories identify the child's decisions, including supported choices, not the adult's hand movements.

No player is the permanent helper

Every role carries meaningful authority and every child may rotate, refuse or combine roles.

Observer

Notifies support, changes or conflicts and may call STOP.

Builder

Moves or selects candidate rule objects through any access route.

Checker

Asks WHERE, MATCH and READY questions.

Story keeper

Records or replays committed changes.

Route guide

Describes topology, landmarks or sequence.

Access captain

Checks pace, quiet, stop and route choices - without controlling others.

- A role may be shared by two children.
- The quietest or slowest role is not automatically lesser.
- A child may observe before joining and still influence the game.
- Cooperative timing waits for all required confirmations.
- The group cannot override an individual STOP.

Communication breakdown repair plan

A repair plan protects the child from repeated guessing and protects the authoritative game state from an unconfirmed action.

1 Freeze

do not commit the uncertain action

2 Acknowledge

say that the interpretation is unclear

3 Restore

return to the last confirmed focus and choice set

4 Re-present

use a different channel or fewer options

5 Confirm

ask using the agreed yes/no/not-sure route

6 Document

update the protocol only with child permission

Never conceal the error

The partner does not pretend a guess was the child's decision.

Never punish the repair

No lost turn, reduced score or public warning follows.

Keep an uncertainty state

The system can hold "unclear" without collapsing it into yes or no.

Learn cautiously

One corrected interpretation does not permanently redefine a body movement.

Authoritative boundary

Only a confirmed child intent creates a proposal. Ambiguous body data may support observation, but it is never an unchecked state write.

Design for functions, not diagnosis labels

Children do not arrive as single-category personas. Design around what must be perceived, understood, operated, communicated, regulated and sustained in this moment.

Functional access model

Access dimension	Questions for the design
Perception	Can essential information be received visually, auditorily, tactually or through a concise partner summary?
Understanding	Is the purpose clear? Are steps, relationships and consequences explicit?
Action	Can the child choose without drag, hold, precision, simultaneous input or fast repetition?
Communication	Can the child stop, refuse, ask, confirm, repair and explain through their own system?
Regulation	Can sound, motion, clutter, vibration, social demand and unpredictability be reduced?
Energy	Can the activity pause, save, shorten and resume without losing concept or status?
Support	Can a partner assist transparently without becoming the decision-maker?

Access envelope

An essential action is complete only when the child can perceive it, understand it, operate it, confirm it, repair it and resume after interruption through at least one working route.

Blind and low-vision access

The game exposes the rule graph, not merely a description of pixels.

Semantic names

Every object has a concise name, role, state, location, relationships and available actions.

Landmark grammar

Rooms have stable regions: entrance, workbench, candidate area, real-world area and exit.

Nonvisual focus

Keyboard, controller, switch and screen-reader focus move predictably and restore after changes.

Zoom and reflow

Text scales, panels reflow, and important content remains usable without two-dimensional scrolling.

Contrast and clutter

Strong contrast, adjustable outlines, no color-only meaning, reduced background detail and optional focus isolation.

Tactile parity

Key rooms and rule objects can be represented with tactile pieces or object cues when useful.

Do not narrate pixels

Describe relationships and choices: “triangle plug connects to the left bridge socket,” not “there is a blue thing near the middle.”

Deafblind access

A deafblind route combines tactile orientation, communication, anticipation and consistent interaction partners. It is not simply screen reader plus vibration.

Tactile home

A stable corner, mat edge or body reference establishes orientation.

Object cue

A real or miniaturized object announces the activity and next transition.

Tactile schedule

A left-to-right or top-to-bottom sequence shows first, next, break and finished.

Touch-consent signal

The child has an agreed way to accept, pause or refuse touch.

Conversation pace

Adults wait, observe, respond and leave space for the child's turn.

Tactile replay

Committed changes are represented by ordered objects, beads or tactile symbols.

Specialist boundary

Individual deafblind communication and tactile-symbol systems require collaboration with the child, family and trained specialists. This document supplies scaffolding, not a clinical prescription.

Deaf and hard-of-hearing access

Speech, sound effects, music and rhythm can enrich play but cannot be the only carrier of rules or warnings.

Captions

Tutorials, dialogue, narration and meaningful non-speech sounds have accurate captions with speaker or source labels.

Visual and tactile cues

Sound events may also appear as icons, safe edge signals, controller vibration or object movement.

Sign-language space

Allow signed video, interpreters or peer signing without covering gameplay or captions.

Conversation control

Text chat, preset phrases, AAC and turn-taking alternatives are available in co-play.

No audio timing trap

Rhythm or direction cues have visible or tactile timing and a practice mode.

Audio customization

Separate music, effects, voice, ambience and accessibility cues.

Caption quality

Captions are synchronized, complete, readable, interruptible and distinguish speakers and meaningful sound sources.

Motor access

No essential action depends on precise pointing, rapid repetition, simultaneous buttons, sustained pressure, twisting, shaking or two-hand movement.

Remap everything

Named actions can be assigned to keyboard, controller, touch, switch, gaze, voice, AAC or partner input.

Toggle alternatives

Every hold has a toggle or timed alternative; every drag has select-then-place.

One-handed play

All actions fit one side of a controller or keyboard and can be sequential rather than simultaneous.

Large forgiving targets

Targets have generous size, spacing, snap and cancellation.

Low-force operation

Digital and physical controls avoid unnecessary force and long reach.

Position independence

Menus and play remain usable seated, lying, mounted, at a desk or with a supporter.

Error recovery

A tremor, spasm, accidental activation or tracking drift is reversible and does not erase the child's work.

Switch access is a complete route

Switch access reaches every essential action, explanation, settings page, puzzle object, replay and exit.

Switch access modes

Mode	Use	Required safeguards
Auto scan	Focus moves on a timer.	Adjustable speed, pause, group size, cue channel and rest item.
Step scan	Child advances focus.	No timeout; back, stop and confirm available.
Row-column	Select row then item.	Stable grid, cancel, spoken or tactile row names.
Group-item	Choose semantic group then object.	Groups match the rule graph, not arbitrary screen position.
Two-switch	Next and choose, or left and right.	One-switch fallback and no simultaneous press.
Partner scan	Person presents choices.	Documented order, wait time, confirmation and fidelity check.

Acceptance gate

A one-switch user can start, change settings, solve a representative room, receive the full ending and exit without a second person making choices.

Eye-gaze and head-pointing access

Gaze interfaces distinguish exploration, reading, rest, scanning and commitment.

Rest and off-screen zones

Provide safe places to look without selection and a way to pause dwell.

Progressive dwell

Show time remaining through at least two channels; let the child cancel by looking away.

Target stability

Do not move targets during selection or hide them behind animation.

Calibration resilience

Support quick recalibration, enlarged targets, smoothing and head pointing.

Critical confirmation

Commit, delete, exit and social actions use a second confirmation route.

Fatigue support

Offer shorter scan sets, dwell-free switch confirmation and frequent save points.

No performance pressure

Gaze speed and accuracy are access characteristics, not scores. The game does not reward “cleaner” eye movements.

Speech, AAC and nonspeaking access

A child may use AAC for conversation, direct switches for gameplay, gaze for navigation and a partner for physical movement - all in one session.

Named intents

The game accepts semantic actions such as STOP, CHOOSE, PREVIEW, CHECK, UNDO and EXPLAIN rather than exact spoken phrases.

Speech optional

Voice control is never required and is not used as an identity test.

AAC bridge

External AAC may activate named actions without losing the child's own vocabulary or voice output.

Conversation presets

Useful phrases support co-play without limiting free communication.

Latency tolerance

The game waits for message construction and does not time out a turn.

Privacy

Do not record raw speech, AAC messages or communication videos unless the child and guardian choose a bounded purpose.

Cognitive and intellectual disability access

The interface reveals purpose, relationships and consequences without relying on memory, inference or abstract metaphor alone.

Clear purpose

Each page or room states what can be changed and why.

Concrete choices

Show real options with real consequences; avoid trick questions.

Small visible steps

Keep the full plan available while presenting one step at a time.

Consistent words and places

Use the same label, symbol and location for the same action.

Examples before assessment

Model a complete try-preview-check cycle without scoring.

Undo and history

Let the child see, repeat and reverse prior actions.

No memory login

Avoid passwords, time-limited codes and repeated personal data when not essential.

Supported explanation

Accept objects, symbols, choice cards, demonstrations and partner-relayed responses.

Reading and dyslexia access

Reading level never decides whether a child can play the rule.

Plain language first

Short sentences, familiar words, active voice and one instruction per step.

Readable typography

Large adjustable text, generous line spacing, left alignment and no dense blocks of italics, all caps or justified text.

Read aloud

Every instruction and label has optional speech with synchronized highlighting when useful.

Symbol plus word

Icons support labels but do not replace them or assume universal meaning.

Mathematical speech

Equations receive a plain-language description and structured reading order.

No text-only puzzle

The same rule can be demonstrated, spoken, touched or acted out.

Choice, not prescription

Do not force a special font. Let the child select font, spacing, size, width and background, then test the real result.

Neurodivergent and executive-function access

Support planning, attention, transitions and regulation without treating harmless movement, repetition or deep interest as problems.

Predictable structure

Show first, next, break and finished; announce changes before they occur.

Focus mode

Reduce visible objects, background motion, particles and optional social prompts.

Task resumption

Keep a visible marker for current step, candidate and last confirmed action.

Interruptions off

Pause pop-ups, reminders and nonessential dialogue during a decision.

Interest-based examples

Allow preferred themes and repeated rooms without reducing challenge.

Flexible repetition

Repeat instructions or actions without nagging, escalating sound or reward loss.

No forced eye contact

Gaze toward a person or camera is never required for participation.

Stimming accepted

Only address movement when there is a real safety or access need.

Sensory and regulation access

The child controls intensity before entering a room and while playing.

Regulation controls

Channel	Controls
Visual	static mode, reduced particles, no glow, softer contrast option, focus isolation, dark and light themes
Motion	reduced motion, fixed camera, slower transitions, no camera shake, no forced parallax
Audio	separate sliders, quiet preset, no sudden peaks, captions, visual or tactile alternatives
Haptic	off, strength, pattern, duration, side and alternative cue
Social	solo, observe first, turn preview, preset phrases, no voice requirement
Environment	micro-session, pause anywhere, predictable loading, no surprise restart

Immediate safe pause

STOP freezes nonessential sensory output first, then offers save, quiet, route change or finish. It does not open a loud confirmation dialog.

Emotional and trauma-informed safety

A game for children does not use fear, humiliation, coercion or uncertainty about adult control as learning mechanics.

Content forecast

Describe potentially intense sound, darkness, chase, conflict, touch or social themes before play.

No punitive failure

Avoid shaming voices, lost streaks, public ranking or irreversible loss after an access error.

Private mistakes

Do not broadcast failed checks or communication repairs to peers.

Choice during help

A supporter asks before taking over a control, moving a body part or revealing a solution.

Safe exit

The child can leave a room or session without confrontation or explanation.

Repair after distress

Restore control, explain what happened, offer quiet, and let the child choose whether to continue.

Safeguarding boundary

This design supports emotional safety but does not replace safeguarding policy, trained staff, clinical care or emergency procedures.

Fatigue, pain, chronic illness and medical variability

A child may have different stamina, strength, attention, vision, hearing, pain or medication effects across a day.

Micro-sessions

Every room has meaningful 2-, 5- and 10-minute stopping points.

Save anywhere practical

Candidate and committed state are preserved separately.

Effort forecast

Before a room, show duration, number of choices, sound, motion, touch and physical reach.

Posture changes

Allow seated, supported, reclined and mounted use without losing orientation.

Low-energy mode

Reduce repetitions and navigation, not conceptual content.

Return summary

Resume with a short context reconstruction rather than a quiz.

No attendance streak

Do not punish missed days or incomplete sessions.

Device handover

Switch between touch, switch, gaze and partner input without restarting.

Photosensitivity, migraine and vestibular access

Some visual and motion effects are safety hazards, not merely style choices.

Flash safety

Do not use rapid flashing. Test content against recognized flash thresholds and prefer a no-flash design.

Motion reduction

Disable camera shake, zoom pulses, parallax, auto-rotation and nonessential interaction-triggered motion.

Pattern control

Reduce high-contrast stripes, dense repeating patterns, flicker and large-area strobing.

Static alternatives

Every animated explanation has a stepwise or still version.

Brightness control

Support dark and light themes, dimming and removal of bloom or glare.

Warning before exposure

Forecast motion and intense visual effects before they appear, with a safe path to settings.

Non-interference rule

A hazardous effect cannot be excused because the child could turn it off after it begins. Safe defaults and pre-entry controls are required.

Change route without losing the world

A route switch is an adapter change, not a new game state.

1 Pause

freeze commit and nonessential sensory output

2 Preserve

store focus, candidate, scan group and explanation

3 Switch

activate the new input or communication route

4 Reorient

summarize location and available actions

5 Confirm

let the child verify restored focus

6 Continue

resume with the same puzzle and reward

Tracking loss

Pause rather than selecting the last gaze target.

Switch disconnect

Keep scan focus and offer keyboard, touch or partner continuation.

AAC unavailable

Provide a low-tech core board and yes/no/stop route.

Sensory overload

Apply quiet or still mode without resetting progress.

Fatigue change

Move to shorter scan groups or partner relay while preserving child confirmation.

Device change

Carry semantic focus and replay, not device-specific coordinates.

Acceptance gate

A representative room can start with touch, continue with one switch, be explained by screen reader and finish by partner scan without state divergence.

Combined-access test cards

Testing includes routes that interact. A design that works for each category separately may still fail when they are combined.

Minimum combined-access scenarios

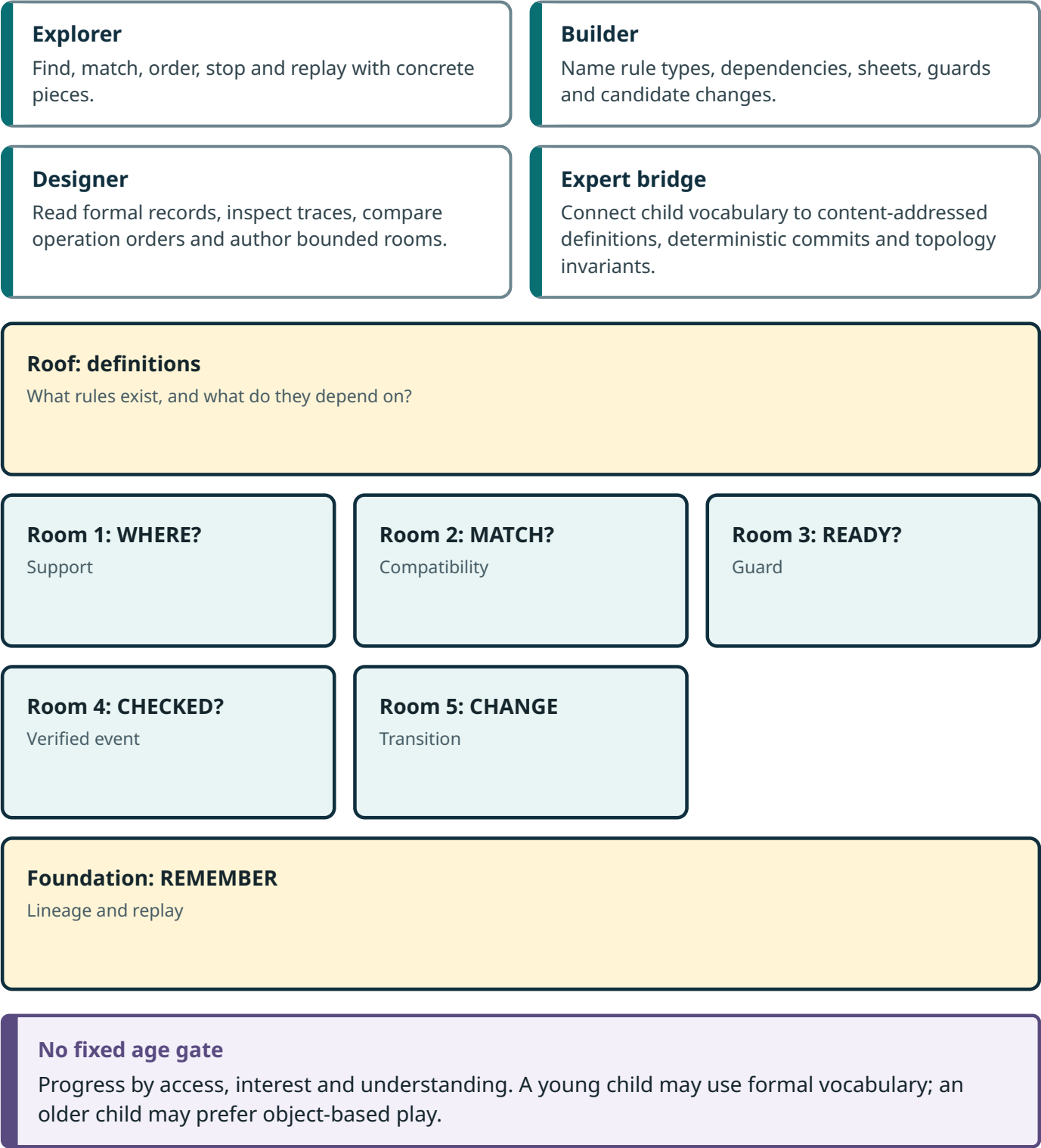
Test combination	Representative task	Pass condition
Screen reader + one switch	Navigate, inspect, preview, commit and undo.	Focus, speech and scan order remain synchronized.
Low vision + eye gaze	Select dense rule graph at 200% text.	Targets remain stable, separated and visible without clipping.
AAC + partner scan	Reject an interpretation and choose another rule.	Partner relays only confirmed intents; NOT THAT works.
Deafblind + fatigue	Use object cue, tactile schedule and a break.	Return restores orientation without forced touch.
Reduced motion + nonvisual	Complete topology room.	No essential route cue depends on animation or stereo sound.
One hand + cognitive support	Reorder three operations.	Select-then-place, step list and undo preserve full concept.
Pain fluctuation + multiplayer	Switch to partner relay mid-turn.	Ownership and credit stay with the child.

Testing rule

A passing route is completed by children who use it, with their preferred supports, not by a nondisabled tester simulating an impairment for a few minutes.

The whole substrate as a learning house

Children encounter the real architecture through objects, questions, stories and play before formal notation.



The six big questions

The questions are short enough for AAC, switch scanning, tactile cards and repeated group routines.

WHERE?

Can this rule reach that object, room, time or route?

MATCH?

Are the types, sheets, phases, roles and policies allowed to work together?

READY?

Has the needed boundary, timing or confirmation condition been met?

CHECKED?

Did the complete proposal pass support, match, guard, confidence and conflict checks?

CHANGE

What exact values, links, routes or ownership records change now?

REMEMBER

What event, reason, sequence and lineage must be kept for replay and return?

WHERE -> MATCH -> READY -> CHECKED -> CHANGE -> REMEMBER

This order stays stable in digital, tabletop, tactile and classroom versions.

Substrate basis

The canonical query-first sequence is retained from the GPU-native addendum and the later game-runtime profiles: support, compatibility, guard, verified event, transition and lineage.

Definitions are physical rule cards

UGTS-KC 3.6 makes definitions first-class records. Definition Breaker turns those records into inspectable and manipulable learning objects.

Definition-object bridge

Child sees	Builder sees	Formal record
A rule card with a name and sockets	Stable rule ID, kind and connections	id, kind, domain, codomain
A picture or tactile shape	Parameters and capabilities	literal parameters and capability flags
Arrows to other cards	Dependencies	dependency IDs
A place on the order rail	Evaluation phase	phase and explicit schedule
A “must stay true” mark	Invariant	declared invariant set
A seal or fingerprint	Content verification	canonical content hash
A story tag	Source class	provenance and evidence boundary

Bounded self-reference

Definitions may describe other definitions, but unknown references, duplicate IDs, invalid hashes and undeclared cycles are rejected before play.

Dependencies say what a rule needs

A rule may depend on shapes, maps, guards or schedules. The game exposes those links instead of hiding them in code.

Needed first

A bridge-turn rule may need a pivot, angle and object identity.

Shared source

Two rooms may reference the same definition without becoming the same instance.

Missing link

A proposal cannot be checked if a required definition is absent.

Cycle

A rule that requires itself through an undeclared loop is rejected or given a bounded cycle policy.

Explain path

The child can ask “What does this need?” and follow one dependency at a time.

Tactile graph

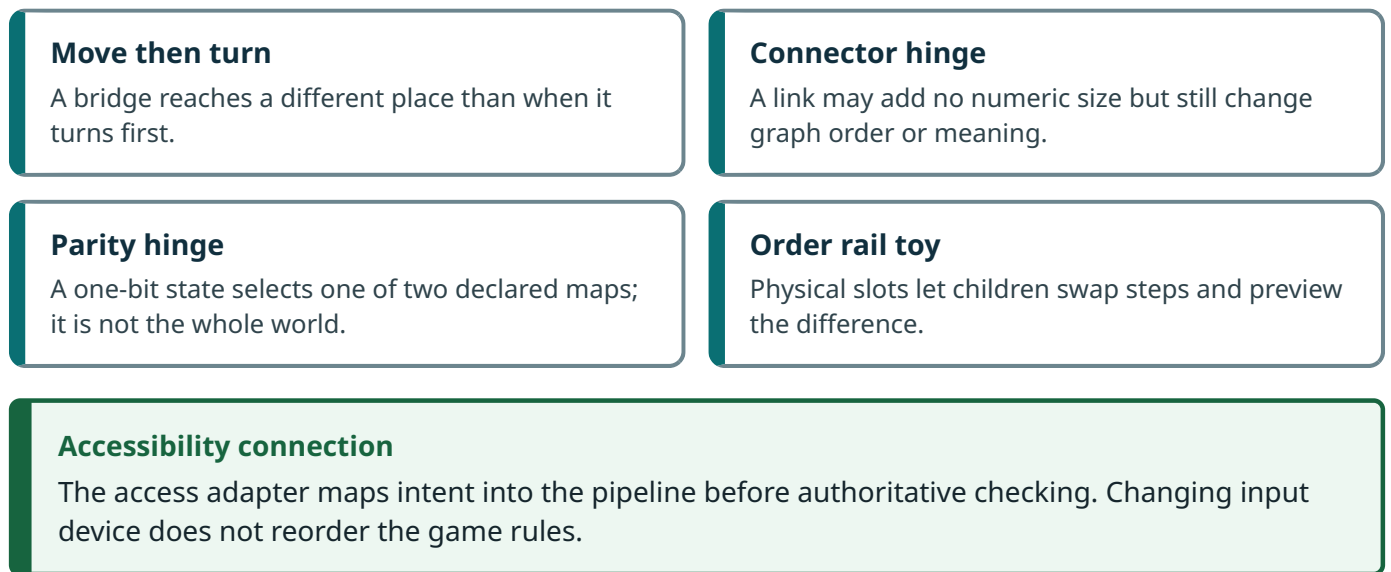
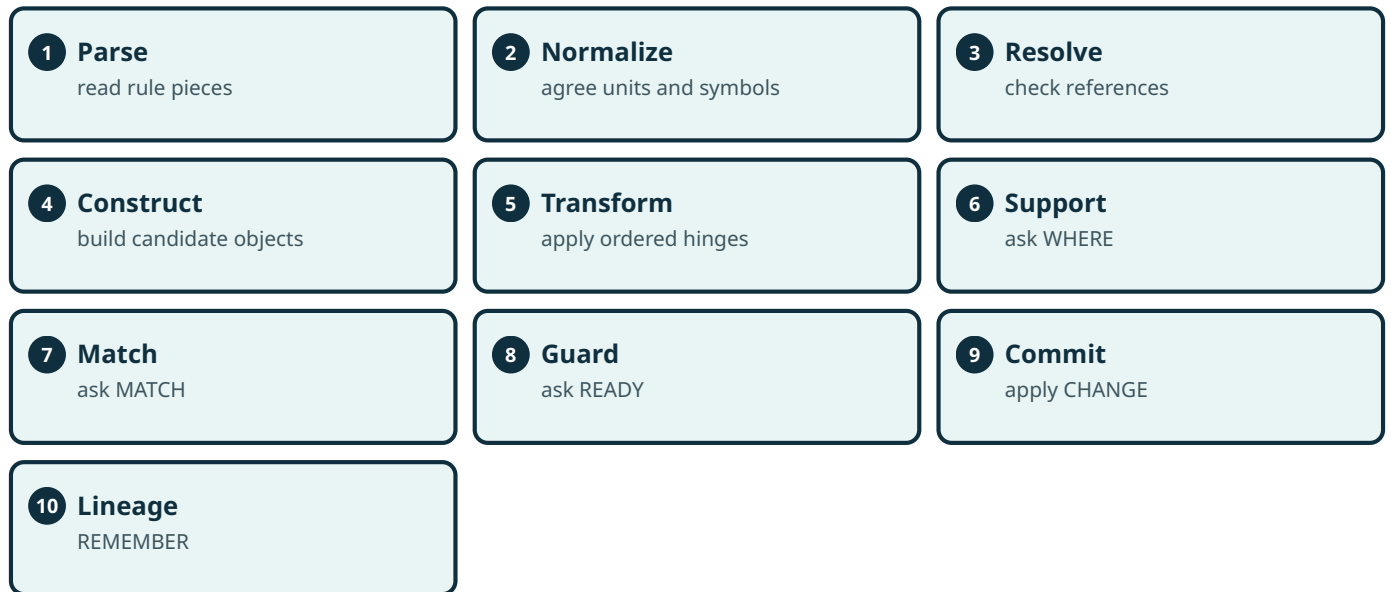
Cables, pegs, raised arrows or adjacency cards can represent dependency edges.

Learning move

Start with “this card needs that card.” Later introduce directed graphs, topological ordering and content-addressed references.

Order changes the world

UGTS-KC 3.6 stores operation order because transforms and rules generally do not commute.



WHERE? means support

Support asks whether a rule or object can reach the current query. It removes irrelevant possibilities without deciding whether a change is allowed.

Room support

Is the object in this room, layer or interest area?

Shape support

Is it inside the cone, shell, sphere, path band or tile region?

Time support

Is the rule active during this interval?

Interaction support

Can the child's current focus, pointer or hand reach this object?

Physical support

Is the rule block on the active mat or connected track?

Access support

Can the current route present and operate the object?

Boundary

Support says "relevant," not "compatible," "safe" or "true." A supported object may still fail later checks.

MATCH? means compatibility

Compatibility checks type, sheet, phase, role, ownership, policy and other declared tags.

Type match

A triangle plug fits a triangle socket; a circle may require an adapter.

Sheet match

Two objects at the same coordinate may be on different topological sheets.

Phase match

A gate may accept only the active phase or rhythm.

Role match

A child may move a shared candidate but not another child's private communication control.

Policy match

The room may allow preview but require joint confirmation for commit.

Access match

The output has a perceivable route and the action has an operable route.

No forced coupling

Co-location, visual overlap, matching color or adult expectation never overrides an explicit incompatibility.

READY? means a guard was checked

A guard is a threshold, crossing, interval, timing or confirmation condition. It carries a margin and a status.

Guard statuses

Child word	Meaning	Example
Before	The condition is not yet met.	The key has not crossed the safe line.
Ready	The accepted condition is met.	The bridge angle is inside its allowed band.
Touching	The boundary is reached without a certified crossing.	A slider rests exactly on a marker.
Unclear	Error or uncertainty overlaps the boundary.	Tracking drift makes the choice ambiguous.
Too many	Repeated crossings would chatter.	A shaking sensor toggles rapidly.
Conflict	Two ready events request incompatible changes.	Two players commit different owners at the same time.

Access guard

Important actions use clear confirmation and accidental-activation repair. A slow or unconventional movement is not rejected merely because it differs from a default gesture.

CHECKED? means a verified event

Input devices, game systems and physical players may propose changes. Only a verified proposal can change the authoritative world.

Proposal

What the child or system wants to change.

Reason codes

Which check passed, failed or remains unclear.

Preview

A reversible candidate world shown in the child's route.

Conflict policy

Priority, merge, shared confirmation or explicit rejection.

Verification

One ordered decision boundary for all input methods.

No unchecked write

A renderer, sensor, AI helper or adult partner cannot silently mutate gameplay state.

```
verified = support_ok
           and compatibility_ok
           and guard_status is accepted
           and confidence is sufficient
           and numeric_error is within the event margin
           and conflict policy is resolved
```


CHANGE is an atomic transition

A transition applies an exact patch to scene, topology, dynamics, ownership or rule state.

Scene patch

Move, rotate, connect, hide or reveal a stable node.

Topology patch

Change sheet, orientation, port or gluing state.

Rule patch

Replace a definition reference or order slot.

Ownership patch

Give, share, hand over or release control.

Dynamic reset

Change mode, velocity, constraint or timer after a verified event.

Access independence

The same patch results whether intent arrived by touch, switch, gaze, AAC or partner relay.

Atomicity

Either the whole accepted patch applies with its lineage record, or none of it does. Partial hidden changes make replay and learning unreliable.

REMEMBER means lineage and replay

Lineage records how a state came to be. It lets the child explain, undo, resume, compare and detect divergence.

Stable identity

An object keeps its ID even when it moves, changes parent or swaps appearance.

Event sequence

Each commit has an order, time, reason and pre/post state check.

Checkpoint

A saved snapshot provides a safe return point.

Replay

Reapply events and stop at the first mismatch.

Branch history

Repeated coordinates can remain distinct because route, winding, sheet or branch differ.

Child explanation

Replay can be one sentence, tactile beads, symbols, full trace or a side-by-side candidate comparison.

Privacy split

Access preferences and communication data are not part of public game lineage. Store only the minimum needed to restore access.

The same place can hold different states

UGTS distinguishes coordinate equality from identity and compatibility. This becomes a powerful child-friendly puzzle idea.

Two tactile sheets

Pieces can appear in the same place but remain on different layers.

Orientation flip

A topological wrap may reverse left and right without applying a physical force.

Winding count

Going around a loop three times is not the same history as going around once.

Route class

Two paths meet visually but approach from different allowed routes.

Compatibility window

Objects couple only when sheet, phase or role tags match.

Accessible representation

Use layer words, tactile overlays, distinct textures, sound labels and explicit screen-reader states.

Do not rely on transparency

Every sheet distinction also exists in words, structure, tactile marks or another nonvisual channel.

Topology tracks make gluing physical

Physical tracks and digital maps can teach non-orientable routing without pretending topology creates energy or material magic.

Topology-track vocabulary

Track element	Child experience	Formal idea
Ordinary portal	Exit at the paired doorway.	Coordinate and port map
Half-turn state wrap	Route turns and an internal marker flips.	Bundle twist with state reversal
Reflective wrap	The return path reverses orientation.	Orientation-reversing gluing map
Sheet bridge	Move from layer A to layer B.	Sheet transition
Winding counter	Add one bead per complete loop.	Phase plus winding lineage
Compatibility gate	Only matching route tokens share the exit.	Route-class compatibility

Evidence boundary

Topology is represented as explicit routing and identity rules. Physical effects still require independent mechanics and measurement.

Patterns and fields are rule families

UGTS-KC 2.0 supplies parametric curves, implicit fields, surfaces, topology descriptors, kinematics and hybrid events. Children can explore these through safe, bounded examples.

Pattern card

A versioned shape or path with parameters, domain, frame and derivative contract.

Field card

A scalar rule that classifies inside, boundary and outside.

Path card

A route with speed, curvature, winding and frame information.

Guard card

A threshold or zero relation promoted to event semantics.

Error card

A sample or approximation carries a tolerance; visual accuracy is not automatically collision accuracy.

Role card

The same pattern may act as art, support, trajectory, guard or calibration path, but the role is explicit.

Child activity

Change one parameter at a time, predict what stays the same, preview the result, and compare the formal definition after play.

Two hands become one accessible transform

KC Two Hands 3.0 defines midpoint translation, separation scale, pair-axis alignment, twist, cooperative ownership and handover. Definition Breaker teaches the idea while providing sequential alternatives.

Bimanual concept parity

Two-hand idea	Physical or digital experience	Accessible equivalent
Midpoint translation	Move an object between two hands.	Choose anchor, then choose destination.
Separation scale	Hands move apart or together.	Select larger or smaller in steps.
Pair-axis alignment	Turn the line between hands.	Choose orientation from a compass or scan list.
Wrist twist	Rotate around the pair axis.	Use clockwise/counterclockwise actions.
Cooperative ownership	Two players hold one object.	Shared confirmation with independent stop.
Handover	One hand releases; the other remains owner.	Explicit GIVE / KEEP / SHARE transition.

No forced two-hand input

The mathematical transform may be learned through two hands, one hand, switches, gaze, buttons, partner relay or physical tiles. The authoritative result stays the same.

Seeds and bounded grammar build large worlds

SCLP 3.6.2 adds deterministic branch bits, finite grammar budgets, winding and packed spatial keys. Definition Breaker uses them as inspectable construction rules.

Seed

A compact starting value selects a reproducible world family.

Branch bit

Chooses one of two declared productions; it is not the whole state.

Grammar budget

Maximum depth, symbols and stack prevent runaway growth.

Address

A finite key locates a quantized cell or state family.

Reconstruction

Store seed plus real novelty when the world can be rebuilt accurately enough.

Kill criterion

Stop when grammar growth, error, collisions or metadata exceed the value of reconstruction.

Child language

A small recipe can build a big world, but the recipe has limits and must remember surprises.

Compression is a promise with conditions

The substrate may reduce storage by referencing definitions, packing state and logging novelty, but width ratios are not automatically equal-information compression.

Compression teaching questions

Question	Child version	Formal check
Can we rebuild it?	Does the seed and recipe make the same world?	Reconstructibility and deterministic profile
What was lost?	Did we forget shape, timing, sheet or uncertainty?	Semantic and error equivalence
Is the key finite?	How many different addresses really exist?	Declared bit capacity and quantization
Are rare changes stored?	Did we remember surprises?	Novelty or event-log completeness
Is it actually cheaper?	Did the smaller box take more work?	Decode, verify, metadata, latency and energy
When do we stop?	Is the shortcut still helping?	Application kill criteria

No magic claim

One bit can store a route or predicate; it cannot replace continuous geometry, uncertainty, policy and lineage.

Public, private and permitted

SARA 3.6.3 is not a children's wallet game. Its suitable contribution is a clear lesson that public information, private information and authorization are distinct types.

Public

A rule name, public test vector, route label or shared game seed may be intentionally visible.

Private

Access settings, communication data, personal identity and secret credentials stay protected.

Permission

Seeing an object does not grant permission to change, search or use it.

Authorization guard

A requested action must fit an allowed scope and avoid forbidden operations.

Public-only certificate

A game may report what was checked without revealing private data.

Child version

"You may see the door number. That does not mean you may enter, copy the key or tell everyone who lives there."

Security boundary

No mnemonic, private-key, wallet-search, signing, funds-transfer or target-address mechanic belongs in ordinary child play. SARA remains an optional privacy-and-permission lesson for older learners.

Three languages, one idea

The words can grow while the underlying relation stays stable.

Stable vocabulary bridge

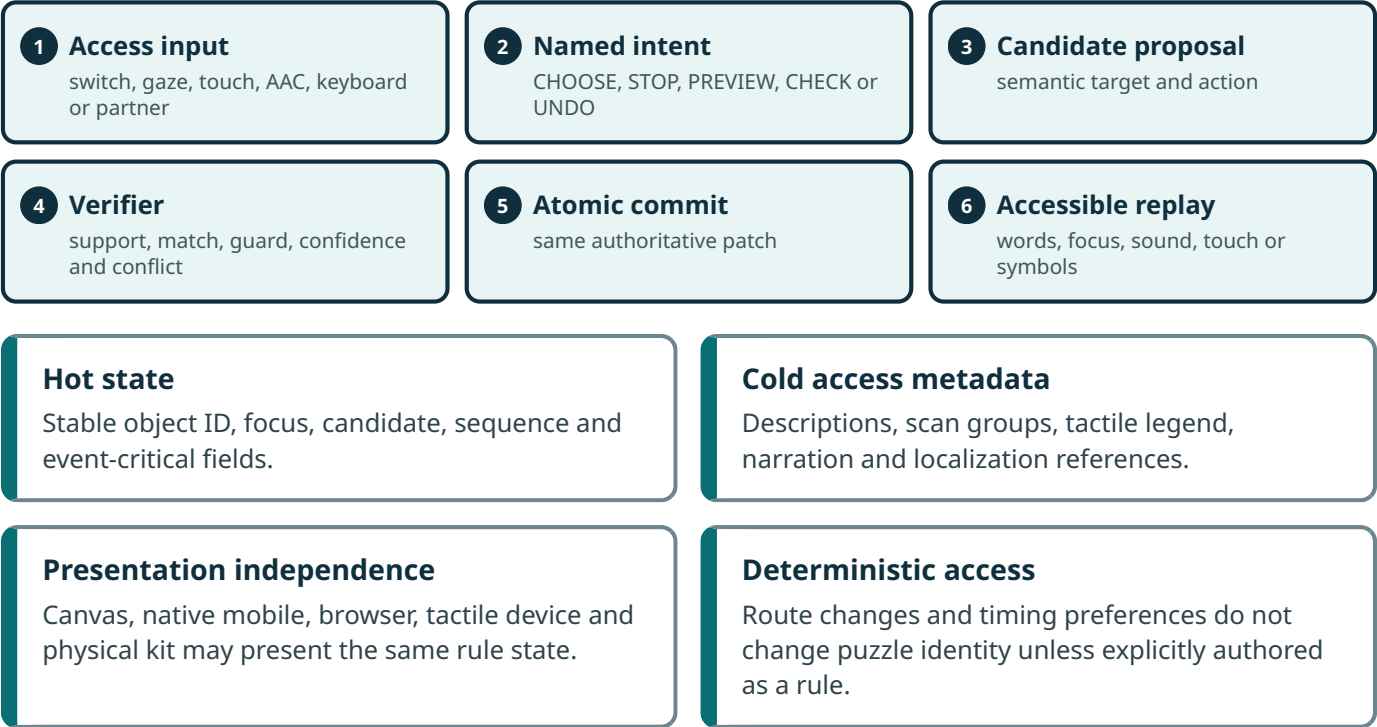
Child question	Facilitator prompt	Formal substrate term
WHERE?	Can this rule reach the object or place?	Support predicate
MATCH?	Are these types, sheets, roles and policies allowed to work together?	Compatibility predicate
READY?	Has the boundary, threshold or confirmation condition been met?	Guard classification
CHECKED?	Did the proposal pass every required check?	Verified event or proposal
CHANGE	What exact part of the world changes now?	Atomic transition or patch
REMEMBER	What must be replayed so we can explain or return?	Lineage, event log and checkpoint
WHAT DOES IT NEED?	Which rules must exist first?	Dependency graph
WHICH STEP FIRST?	Does order change the result?	Explicit phase and composition order

Learning rule

Never introduce a child-only metaphor that cannot later connect to the formal term. A gate can begin as READY? and grow into interval guards, margins, uncertainty and event ordering.

Digital child-access architecture

Accessibility adapters propose the same named intents to the same deterministic verification boundary. They do not create parallel game logic.



Every rule object needs a semantic contract

A picture, sound or tactile token is a presentation of an object. Assistive routes need the underlying meaning.

Minimum semantic fields

Field	Required content
name	Short child-facing label that remains stable.
role	Rule, object, connector, guard, route, history, control or status.
state	Current value, selected, disabled, uncertain or committed status.
location	Room, region and relative landmark; screen coordinate is optional.
relationships	Dependencies, sockets, owner, parent, route, conflicts and affected targets.
actions	Named available actions with consequences and confirmation requirements.
description_short	One-sentence summary.
description_detail	Parameters, invariants, first failed check and replay history.
access cues	Caption, speech, scan group, tactile mark, earcon and haptic mapping.
privacy	Whether the field is public, local, temporary or prohibited from logs.

Authoring gate

A decorative asset may be hidden from assistive technology. Any object that affects decisions, progress or story has complete semantics.

AccessProfile is private preference, not puzzle state

The profile stores what usually helps. It never changes score, challenge identity, authoritative replay or public ranking.

Local first

Store on device or in an explicitly chosen private account area.

Minimal data

Keep settings, not diagnosis, raw gaze, speech, video or inferred emotion.

Child editable

The child can view, change, reset or temporarily override preferences.

Portable export

A plain-language structured file may move settings between trusted devices with consent.

No paywall

All accessibility settings are included and available offline where practical.

No public badge

Access choices are never displayed as achievement, deficit or matchmaking category.

```
AccessProfile = {  
  perception_routes, action_routes, communication_routes,  
  text_and_language, scan_and_dwell, sound_motion_haptic,  
  partner_protocol, privacy_choices, last_reviewed  
}
```

AccessSessionState may change minute by minute

Current fatigue, sensory load, tracking status, partner availability and scan position belong in a short-lived session record.

Separate from profile

A hard day does not rewrite the child's usual settings.

Separate from world

Changing scan speed or quiet mode does not alter puzzle state.

Recoverable

Device loss or pause restores semantic focus and pending confirmation.

Ephemeral

Discard sensitive temporary data when the session ends unless the child asks to save a preference.

Explainable

The child can ask "What route is on?" and receive a plain answer.

No hidden adaptation

Automatic changes are offered and confirmed, not silently imposed.

```
AccessSessionState = {  
  active_routes, focus_id, scan_group, dwell_progress,  
  quiet_level, motion_level, verbosity, partner_state,  
  device_status, pending_confirmation, last_clear_checkpoint  
}
```

Map devices to intent

KC Elizabeth 3.9 already uses named multi-device actions. DB-1.0 extends the principle to assistive and communication routes.

Named intent map

Named action	Possible bindings
FOCUS_NEXT / PREVIOUS	keyboard, controller, switch, scan, gaze region, partner
CHOOSE	key, button, touch, dwell, AAC intent, yes signal
PREVIEW	key, switch item, spoken command, partner card
CHECK	button, switch, gaze target, AAC phrase
UNDO / REPAIR	dedicated button, scan item, NOT THAT signal
STOP	escape, controller, touch corner, switch pattern, AAC, partner stop card
DETAIL_MORE / LESS	keys, rotor, scan group, speech or AAC
CHANGE_ROUTE	settings shortcut, scan item, AAC, supporter with confirmation

Adapter rule

Adapters normalize device events into intents. They never bypass support, compatibility, guard, verification or lineage.

Scanning is a state machine, not a timer trick

Scanning is deterministic, inspectable, interruptible and recoverable.

Stable order

Use semantic groups and preserve order across repeats.

Adaptive only by consent

The system may suggest slower scanning or smaller groups but does not silently reorder choices.

Rest item

A neutral focus stops activation pressure and permits regulation.

Recover focus

After pause, dialog or device loss, restore the same object and group.

Multiple cue channels

Focus may be visual, spoken, tactile or haptic with synchronized timing.

Repair path

STOP, BACK and NOT THAT remain in every reachable scan state.

```
scan = (mode, ordered_groups, current_group, current_item,  
        cue_channel, interval, pause_policy, confirmation,  
        repeat_policy, stop_item, help_item, repair_state)
```


Gaze interaction is a guarded proposal

Raw gaze is noisy observation. The system promotes it to intent only through declared dwell, confirmation and repair rules.



Screen readers need graph queries

A flat list of buttons cannot explain a rule world. The runtime exposes structured queries over objects, dependencies, routes and history.

Minimum nonvisual graph queries

Query	Child-facing answer
summarize_scene()	"You are at the Commit Bench. Three rule cards are in the candidate area."
describe_focus()	"Turn rule, selected, depends on Pivot and Angle."
list_neighbors()	"Previous: Move. Next: Scale. Connected socket: Bridge."
list_dependencies()	"Turn needs Pivot and Angle; both are ready."
first_failed_check()	"MATCH failed: circle plug does not fit triangle socket."
route_history()	"You crossed the reflective wrap once; orientation is reversed."
replay_since(checkpoint)	"Two checked changes: connect support, then move bridge."
available_actions()	"Preview, explain, replace, undo or return."

Verbosity control

Each query supports brief, standard and detailed output. The child can interrupt speech and ask for one relation at a time.

Focus and navigation are world state for the child

Focus is presentation state, but losing it can make the world inaccessible. It therefore receives explicit continuity and repair rules.

One active focus

Every operable object has a stable focus ID and a visible, spoken or tactile focus cue.

Logical order

Navigation follows semantic groups and relationships, not drawing or DOM accident.

Dialog return

Closing help, settings or a warning returns to the invoking object.

Commit return

After a transition, focus moves to the changed object or the next named step and announces why.

Deletion repair

If an object disappears, focus moves to its parent, previous neighbor or stable room landmark.

No focus trap

The child can always reach STOP, HELP, settings and exit without restarting.

Automated check

Every interactive object has a unique focus ID, an accessible name, an action, a deterministic next/previous relation and a safe return target.

Text, symbols and language

All essential text is real selectable text or has an equivalent semantic label. Decorative letterforms never carry the only meaning.

Scale and reflow

Support substantial text enlargement without clipped controls or lost relationships.

Measure line length

Use readable line widths and adjustable spacing rather than shrinking text to fit.

Stable terminology

Use one primary label per action and explain synonyms.

Symbol redundancy

Combine shape, border pattern, word and optional sound or tactile mark.

Plain plus formal

Show child wording first and reveal the formal term on request.

Localization

Externalize strings, allow text expansion and provide grammar-aware sentence assembly.

No icon-only safety controls

STOP, HELP, privacy, delete and consent actions always include words or accessible names and are never distinguished by color alone.

Audio, captions and narration

Audio can teach timing, structure and atmosphere, but the child controls it and every essential cue has an equivalent route.

Audio access contract

Audio element	Required access behavior
Narration	Adjust voice, speed, verbosity and interruption; replay any line.
Dialogue	Captions, speaker identification, transcript and pause-on-choice.
Sound effect	Caption or visible/tactile source cue when meaningful.
Music	Independent volume; do not encode an essential rule only in harmony or rhythm.
Earcon	Stable legend; optional visual or tactile equivalent.
Spatial sound	Recenter and describe relative direction; do not require headphones.
Voice input	Optional, privacy-bounded and replaceable by named actions.

Quiet first-run

The first launch does not begin with loud music, surprise speech or motion before STOP and volume controls are reachable.

Sonification and haptics use a legend

A sound or vibration may reinforce the substrate sequence, but it never becomes a secret code required for progress.

Illustrative multisensory legend

Rule state	Possible sound	Possible haptic	Non-audio/tactile equivalent
WHERE supported	short low tone	one soft pulse	solid border and word SUPPORTED
MATCH compatible	two rising tones	two pulses	matching edge shape and word MATCH
READY guard accepted	held tone resolves	long pulse	gate opens and text READY
UNCLEAR	soft alternating tone	slow alternating pulse	striped border and word UNCLEAR
CHECKED	short chord	firm single pulse	check state and spoken confirmation
REMEMBER	three-note trace	three small pulses	replay token or event number

Customization

Frequency, loudness, rhythm, haptic strength and pattern are adjustable or removable. The child may select a different legend.

Constructive error messages expose the first failed check

A failed proposal returns a typed reason, a child-readable explanation and an accessible repair action.

Constructive rejection examples

Reason	Child explanation	Repair
OUTSIDE_SUPPORT	"This rule cannot reach the bridge from here."	Move the rule, change support, or show the support boundary.
INCOMPATIBLE_TYPE	"The circle plug does not fit the triangle socket."	Choose an adapter or another socket.
GUARD_NOT_READY	"The gate has not crossed its safe line yet."	Continue preview or inspect the margin.
UNCERTAIN_INPUT	"I am not sure which target you chose."	Recalibrate, enlarge, use switch or partner confirm.
CONFLICT	"Two changes want different owners at the same time."	Choose order, share, merge or cancel.
MISSING_DEPENDENCY	"Turn needs a pivot rule first."	Find or create the missing rule.
CYCLE_REJECTED	"These rules would need each other forever."	Break the dependency or add a bounded cycle policy.

Language gate

Never use "invalid user," "bad input," "failed child," or inaccessible numeric codes without explanation.

Save, undo and replay are access infrastructure

Children must be able to explore without the fear that an access error or body movement will destroy work.

Candidate autosave

Preserve the current uncommitted construction separately from the real world.

Frequent checkpoints

Create meaningful return points after each checked change and before intense rooms.

Unlimited undo where safe

Undo returns the authoritative state and explains what was reversed.

Replay by channel

Text, speech, tactile tokens, animation, symbols or partner summary.

First divergence

When replay differs, stop and identify the earliest mismatched event or definition.

Exported learning trace

With consent, save a privacy-safe portfolio of decisions and explanations, not raw access telemetry.

No punishment for recovery

Undo, save, reload, hints, route changes and breaks do not lower rewards or mark a child as assisted.

Accessible multiplayer and co-play

Local and online play preserve individual access routes, private communication and independent STOP.

Per-player access

Each player has separate text, sound, input, scan, motion and communication settings.

Private prompts

A child's screen reader, AAC choices or support instructions do not need to be broadcast.

Turn flexibility

Use untimed turns, ready checks, turn preview and asynchronous responses.

Shared commit

Joint actions show each required confirmation and allow any participant to withdraw.

Role parity

Every role can inspect, propose, confirm or replay something meaningful.

Moderation and safety

Block, mute, report, leave and trusted-adult routes are accessible from every input method.

No forced disclosure

Matchmaking does not require a diagnosis, communication method or accessibility label. Children decide what to share.

Offline and low-end access

KC Elizabeth 3.9 includes self-contained offline HTML5 delivery. DB-1.0 uses that boundary to reduce access barriers and preserve predictable behavior.

Offline first-run

Core settings, tutorials, captions, narration text, printables and representative rooms work without a network.

Low-spec mode

Reduce particles, background art, animation and audio complexity before reducing semantic information.

Small downloads

Reference reusable definitions and vector assets; avoid large video-only instruction packages.

Local saves

Access settings and progress persist without an account.

No online gate

A child can reach the main learning path without email, captcha, social login or subscription.

Update safety

Schema migrations preserve access preferences, focus landmarks and replay compatibility or explain what changed.

Performance accessibility

Stable frame pacing, fast response and low memory use reduce fatigue, tracking errors and sensory unpredictability. Performance is part of access when measured at equal semantics and error.

Privacy and child data

Access can require personal preferences, but the product must not turn those preferences into surveillance.

Privacy defaults

Data class	Default treatment
Access settings	Local, editable, exportable and deletable.
Diagnosis	Not required; do not infer from behavior.
Raw gaze or face video	Do not store by default; process transiently when possible.
Speech or AAC messages	Do not retain for analytics or model training by default.
Partner notes	Store only agreed protocol fields; avoid subjective labels.
Game lineage	Keep world events separate from private access data.
Portfolio evidence	Opt-in, purpose-limited, child-readable and revocable.
Online identity	Minimum necessary; no targeted advertising or public access badge.

Child-readable privacy

The child can ask: “What do you know about me?”, “Why do you need it?”, “Who can see it?”, “Can I change or delete it?”

AI and adaptive-help boundary

An AI helper may explain, summarize or offer options, but it does not infer disability, emotion, intent or consent and does not bypass deterministic game authority.

Allowed help

Rephrase a rule, produce a shorter summary, suggest another access route, generate a practice example or translate approved content.

Human-checkable output

Every suggestion identifies the source rule and can be rejected, undone or compared with the formal trace.

No hidden difficulty change

Adaptive support does not silently lower concept, reward or story.

No emotion diagnosis

Gaze, voice, movement and latency do not become labels such as bored, defiant, anxious or incapable.

No consent inference

The model cannot treat silence, stillness, compliance or predicted preference as permission.

No authoritative mutation

AI output is a proposal. Support, compatibility, guard, conflict and child confirmation still apply.

Child override

STOP, NOT THAT, SHOW SOURCE, LESS HELP and HUMAN HELP remain available whenever adaptive assistance appears.

Accessibility lint before playtest

Automation cannot prove child access, but it can block common failures before they reach children.

Minimum automated lint set

Lint check	Failure condition
Semantic object	Interactive object lacks name, role, state, action or stable ID.
Input route	Essential action requires drag, hold, simultaneous press or precision with no alternative.
Sensory redundancy	Meaning exists only in color, sound, motion, position or transparency.
Focus	Unreachable object, duplicate ID, unstable order, trap or missing return target.
Text	Clipping, fixed tiny size, image-only instruction, low contrast or untranslated string.
Captions	Missing speaker, source, timing, meaningful sound or transcript.
Scanning	STOP or HELP absent from loop; timeout removes an option.
Replay	Committed event lacks reason, accessible description or undo boundary.
Privacy	Raw gaze, speech, diagnosis or partner notes enter public logs.
Parity	Accessible route receives different puzzle, reward or ending without a declared player choice.

Human gate

Passing lint is necessary, not sufficient. Children who use the routes still decide whether the product is accessible, understandable and worth playing.

Digital acceptance matrix

The vertical slice is accepted only when each route reaches the same room goal and authoritative ending.

Target parity matrix

Route	Start	Inspect	Propose	Confirm	Repair	Finish
Keyboard only	YES	YES	YES	YES	YES	YES
Controller one hand	YES	YES	YES	YES	YES	YES
Touch select-then-place	YES	YES	YES	YES	YES	YES
One switch	YES	YES	YES	YES	YES	YES
Eye gaze + dwell	YES	YES	YES	YES	YES	YES
Screen reader	YES	YES	YES	YES	YES	YES
AAC + partner relay	YES	YES	YES	YES	YES	YES
Tactile physical bridge	YES	YES	YES	YES	YES	YES

Evidence requirement

Each YES is supported by a recorded test, child or representative-user feedback, defect log, fix and retest. A design intention is not evidence.

Physical play carries the same rule world

A physical piece is not merely a themed toy. It represents the same stable ID, role, state, relationship and event sequence used digitally.

Physical-digital equivalence

Physical element	Digital equivalent	Shared meaning
Rule tile	Semantic rule object	Definition ID, type, parameters and actions
Socket or edge	Compatibility relation	Which types may connect
Dependency cable	Dependency edge	What must exist first
Order rail	Phase schedule	Which operation occurs first
Try mat	Candidate world	Safe preview without commit
Commit gate	Verifier	Support, compatibility, guard and confirmation
Replay beads	Event log	Ordered checked changes
Sheet overlay	Topology state	Layer, orientation and route identity

Parity rule

A child can begin with a physical piece, continue digitally and return to the physical kit without learning a contradictory vocabulary.

Universal physical-product requirements

Every prototype begins with child agency, low effort and redundant meaning. Product engineering later adds age grading, material certification and regulatory testing.

Large and graspable

Offer multiple sizes, easy-pick edges, finger spaces and tools for pointing or lifting.

Low force

Connections release without painful grip, twisting or sustained pressure.

Stable mounting

Use anti-slip mats, trays, lap boards, vertical boards or clamps selected with the child.

No color-only meaning

Combine shape, border, texture, word, orientation notch and optional sound.

Quiet and magnet-free variants

Provide non-clicking, nonmagnetic alternatives for sensory, medical and safety needs.

Cleanable and inspectable

Pieces can be cleaned, checked for damage and replaced without destroying the rule set.

One-hand and partner routes

Every two-piece action can be staged, fixed, held by a jig or relayed with confirmation.

Stop and break objects

A large STOP card and a tactile break token remain within reach.

Prototype boundary

These are design requirements, not manufacturing drawings, age grading, choking-hazard assessment, material certification or regulatory approval.

A redundant tactile symbol grammar

Every rule class has several identifiers. Children may use any subset that is meaningful to them.

Illustrative symbol grammar

Rule class	Shape	Border	Tactile mark	Short word
Support	rounded rectangle	solid	one raised bar	WHERE
Compatibility	hexagon or keyed edge	paired notch	two matching dots	MATCH
Guard	gate shape	double line	raised threshold	READY
Verified event	shield or check tile	thick frame	central raised point	CHECKED
Transition	arrow tile	direction notch	raised arrow	CHANGE
Lineage	chain or bead tile	linked border	three raised beads	REMEMBER
Dependency	connector tile	open ports	one groove to each port	NEEDS
Uncertain	cloud tile	broken or striped border	wavy texture	UNCLEAR

Test with children

No symbol, texture or shape is universal. Keep stable assignments within a child’s kit and change them when the child finds a clearer distinction.

Rule tiles with typed sockets

Rule tiles make definitions, roles and compatibility physically inspectable.

Front face Large name, rule class, simple example, stable ID and optional tactile mark.	Back face Plain-language definition, formal term, dependencies, invariant and evidence note.
Typed edges Mechanical keys or printed shapes indicate allowed connections without color dependence.	Orientation notch One corner or edge marks top and reading direction.
State sleeve A removable frame shows candidate, checked, committed or uncertain state.	Access handle Optional ring, tab, suction tool point or magnetic-free wand interface.
Prototype tile sizes: pocket 55 mm, tabletop 85 mm, large-access 120 mm. Choose by reach, vision, grasp, group size and mounting.	
Stable identity Changing a tile's position or sleeve does not change its rule ID. Replacing the definition card is an explicit rule transition.	

Support mats and dependency links

A support mat shows where a rule is relevant. Dependency links show what must exist before the rule can be checked.

Support region

A raised or bordered region accepts only objects currently in scope.

Nested support

Concentric mats or trays represent local shells or levels of relevance.

Dependency cord

A low-tension cord, strip or adjacency card connects a rule to what it needs.

Missing dependency flag

An empty port or tactile marker shows that a needed rule is absent.

Cycle check

Children follow links; an undeclared loop receives a STOP/CYCLE marker.

Non-string variant

For children who cannot safely use cords, use side-by-side numbered ports, pegboards or magnetic-free snap tracks.

Teaching sequence

Ask WHERE first: is the piece on the active mat? Then ask NEEDS: are all dependency ports connected? Only then move to MATCH and READY.

Compatibility edges and guard gates

Compatibility and guard are different questions, so they receive different physical mechanisms.

Compatibility and guard mechanisms

Mechanism	What the child does	What it teaches
Keyed edge	Match shape or texture pairs.	Type and policy compatibility
Sheet sleeve	Place matching layer sleeves on both objects.	Same coordinate can be incompatible
Role token	Add owner, shared or private tag.	Permission and ownership compatibility
Threshold gate	Move a slider into an accepted band.	Guard interval and margin
Hysteresis gate	Enter at one mark and leave at another.	Debounce and stable state
Uncertainty band	Use a wide textured area around the threshold.	Not-yet-clear status
Commit latch	Press, point or partner-confirm after checks.	Verified event before transition

No fake fit

Pieces may be physically stackable yet formally incompatible. The explicit tag or socket rule, not accidental physical overlap, decides coupling.

The order rail

A long rail with large slots makes operation order visible and changeable.

1 Choose tiles

MOVE, TURN, SCALE, CONNECT or CHECK

2 Place order

put each tile in one numbered slot

3 Build candidate

apply the sequence on the try mat

4 Compare

swap two tiles and repeat

5 Check

ask WHERE, MATCH and READY

6 Commit

record the accepted order with replay beads

Low-motor route

A partner places each tile only after the child selects a numbered slot.

Nonvisual route

Slots have raised numbers and left-to-right orientation markers; each tile has a spoken and tactile label.

Cognitive route

Show only two tiles first, then add one operation at a time.

Formal bridge

Later write the result as composition and discuss why the operations may not commute.

Core room

The Order Rail remains the recommended first flagship puzzle because it exposes the Definition Breaker premise with minimal reading and no fast timing.

Topology tracks and sheet overlays

Tracks make quotient maps and route history inspectable through movement, touch and tokens.

Ordinary loop

A token returns with the same orientation.

Reflective loop

A keyed token returns with its orientation marker reversed.

Sheet bridge

A sleeve or layer card changes while coordinate remains aligned.

Winding beads

Add one bead each time the token completes the declared loop.

Projected crossing

Two tracks overlap visually but remain on different raised levels.

Compatibility port

A route token couples only at a matching sheet, phase or role gate.

Safety and evidence

Tracks have bounded size, visible endpoints and explicit rules. They do not claim literal infinite surfaces, physical self-intersection or topology-powered energy.

Replay beads and lineage chains

A physical event log lets a child reconstruct what changed and where a branch began.

Lineage-chain vocabulary

Token feature	Meaning
Large bead shape	Event class: connect, move, guard, route, owner or undo.
Raised count	Sequence number or turn group.
Connector direction	Predecessor and successor order.
Branch clip	A split created two possible histories.
Merge ring	Two histories were explicitly reconciled.
Uncertainty spacer	The event was previewed or unclear but not committed.
Checkpoint block	Safe place from which replay can restart.
Private sleeve	Access information is excluded from the public game history.

Accessible replay

The child may touch the chain, hear each event, scan through tokens, watch a digital animation or ask a partner to relay it. The order and meaning remain the same.

Object cues and tactile schedules

Object cues tell the child which activity is coming. Tactile schedules show the sequence and make breaks visible.

Activity cue

A mini rule tile means Definition Breaker; a track segment means topology play.

Start basket

The cue moves from the schedule to a start location when the activity begins.

Finished box

The child places the cue in a finished container to mark closure.

Break token

A distinct object allows a break without ending the whole session.

Change warning

A transition object announces a new room, partner or sensory condition.

Child choice

Offer two object cues when the child can choose the next activity.

Individual meaning

Object cues become meaningful through consistent shared use with the child. A generic miniature is not automatically understood as a symbol.

Low-cost household prototype kit

The first accessibility tests can use ordinary materials while preserving the rule structure.

Household prototype materials

Item	Household prototype	Access variation
Rule tiles	Thick card or foam-board squares	Add tabs, large print, raised glue lines or textured tape.
Sockets	Cut-paper shapes or hook-and-loop patches	Use keyed trays or side-by-side matching cards.
Order rail	Egg carton, ice-cube tray or numbered cups	Mount vertically or on a lap board.
Support mat	Tray, placemat or cardboard border	Add raised edge and home corner.
Guard gate	Clothespin slider or card window	Use low-force pointer or partner movement.
Replay chain	Large beads, blocks or numbered cards	Use rigid links for easier handling.
Sheet overlays	Opaque or translucent folders	Use distinct textures, not transparency alone.
STOP and HELP	Large cards or objects	Keep within the child's easiest reach.

Safety check

Adults inspect size, sharp edges, cords, adhesives, allergens, magnets, batteries and mouthing risks before use. Household prototypes are not automatically toy-safe.

Mounting, reach and low-force operation

A good concept can remain inaccessible if pieces slide, require pinching, sit beyond reach or demand unsupported posture.

Anti-slip base

Use a stable mat, tray or weighted board that does not move during selection.

Vertical option

Mount tiles on a sloped or vertical surface when that improves vision, reach or head control.

Lap and bed use

Provide a lightweight board with edge guards and secure positioning.

Staged action

A piece can be selected now and placed later; a jig holds intermediate state.

Access tools

Pointers, cuffs, handles, suction tools and switch-activated movers may relay a child's choice.

Force budget

Measure connection force, release force, reach distance, repetitions and time rather than saying "easy."

Positioning boundary

Child-specific seating, splinting and positioning decisions belong with the child, family and qualified professionals. The kit adapts to the chosen position.

Rule Repair Crew

Players repair a broken rule world by building a candidate, explaining the first unmet check and committing one safe change.

Rule Repair Crew scaffold

Element	Prototype rule
Goal	Restore three world functions before the event deck ends.
Roles	Builder, Checker, Route Guide, Story Keeper and Access Captain.
Turn	Observe -> choose -> place on TRY -> ask six questions -> confirm -> move to REAL.
Failure	A failed check reveals a repair clue; no player loses a turn.
Communication	Speech, AAC, pointing, switch scan, tactile choice or partner relay.
Difficulty	More dependencies, sheets, guards and conflicts - not faster timing or smaller pieces.
Replay	Add one bead or event card for every committed change.
Win	The group explains the final world through any combination of channels.

Accessibility objective

Every role can stop the game, ask for detail, propose a repair and receive credit. No child is assigned permanent helper status.

Worldseed Builders

A seed card and finite grammar create a shared world. Children decide when reconstruction is enough and which surprises need to be remembered.

Seed card

Selects a starting grammar and style; includes large code, tactile pattern and spoken label.

Production tiles

Replace one symbol with a bounded set of pieces.

Budget track

Shows remaining depth, symbols and branch choices.

Novelty token

Records an external change that the seed cannot reconstruct.

Compression check

Compare “store everything” with “store seed plus surprises.”

Kill criterion

Stop expansion when pieces, time, uncertainty or effort exceed the learning value.

No infinity promise

The world may be large and shareable, but every physical and digital construction has declared finite budgets and addresses.

Dual Authority

Two players control different checks of the same candidate world. Neither can be reduced to an ornamental helper.

Access variation
Each player uses independent communication and input routes.

Solo variation
One child plays both roles in sequence with a visible role token.

Three-player variation
Add a Guard Keeper who controls READY and uncertainty.

Formal bridge
Discuss distributed authority, conflict policy and explicit confirmation.

Dual Authority role split

Player A authority	Player B authority	Shared rule
WHERE and dependency inspection	MATCH and policy inspection	Both may call STOP.
Proposes spatial placement	Proposes sheet, role or ownership tag	Candidate remains reversible.
Confirms support	Confirms compatibility	READY is checked together.
May request more detail	May request another route	Commit requires both agreed signals.
Receives equal replay credit	Receives equal replay credit	Either may withdraw before commit.

Replay Detective

Players reconstruct a world from checkpoints and event tokens, then find the first point where two histories differ.

1 Choose checkpoint

start from a known world

2 Read event

receive it by text, speech, touch or symbol

3 Apply patch

change the physical scene

4 Compare

check the expected state or hash card

5 Stop on difference

identify the first divergence

6 Explain

name the event, rule or order that caused it

No-memory route

Events remain visible; players never have to memorize the sequence.

Nonvisual route

Tactile scene, spoken event cards and raised sequence numbers.

Switch route

Scan through event cards and choose APPLY, EXPLAIN, UNDO or STOP.

Learning bridge

Introduces deterministic replay, pre/post state checks and lineage.

Same Place, Different Sheet

Two objects share a board coordinate but belong to different sheets. The puzzle is to make their compatibility explicit without relying on transparency or color.

Board A raised grid with two distinct sheet overlays or textured sleeves.	Objects Matching coordinates, different sheet and orientation tokens.
Goal Connect the objects through a legal sheet bridge and record the orientation change.	Preview Use a separate try grid or removable sleeves before changing the real board.
Access cues Spoken sheet name, tactile texture, shape code and stable location description.	Replay Add sheet-transition and orientation beads to the lineage chain.
Teaching sentence "Same place" answers WHERE. It does not answer MATCH. Sheet, orientation and policy still matter.	

The Missing Shackle

A maintained relation is represented by a physical constraint bar. Release is one discrete event, not an endless stream of tiny collisions.

Constraint-release scaffold

State	Physical representation	Formal idea
Free	Objects move independently.	No active constraint
Supported	Objects enter the relevant region.	Support true
Coupled	A constraint bar or shared cradle joins them.	Maintained constraint
Release ready	Guard marker reaches accepted band.	Release guard
Released	Bar is removed after confirmation.	Discrete release event
Remembered	A release token enters the replay chain.	Lineage record

Safety boundary

The “shackle” is a conceptual name for a low-force rule bar. It is not worn, locked on a body, tightened, or used as restraint.

Physical prototype evidence sheet

Every prototype records what was tested, who controlled the decisions and where the design created unnecessary effort or risk.

Physical prototype evidence record

Evidence field	Record
Prototype version and date	_____
Children and supporters involved	Use privacy-safe participant codes and access routes.
Purpose of the session	_____
STOP success and time	_____
Meaningful choices completed	_____
Force, reach, repetition and posture issues	_____
Sensory and communication issues	_____
Partner-fidelity observations	_____
Child requested changes	_____
Safety defects and disposition	_____
Retest evidence	_____

No “they managed” acceptance

A child completing a task through strain, repeated prompting, adult substitution or distress is evidence of a barrier, not proof of accessibility.

A twelve-session inclusive learning arc

Each session can be 10 to 45 minutes, split into micro-sessions, repeated or replaced by physical, digital, tactile or acted versions.

Twelve-session arc

Session	Core idea	Accessible activity
1	Control and STOP	Choose a route, make one reversible change and practice STOP or NOT THAT.
2	WHERE	Place rule tiles on support mats or explore digital regions.
3	MATCH	Connect typed sockets, sheets and roles.
4	READY	Use guard gates, uncertainty bands and confirmation.
5	CHECKED	Build a candidate and explain the first failed check.
6	CHANGE	Commit one atomic patch and undo it.
7	REMEMBER	Create a replay chain and return from a checkpoint.
8	Dependencies	Follow NEEDS links and repair a missing definition.
9	Order	Compare move-then-turn with turn-then-move.
10	Topology	Use sheet overlays, reflective tracks and winding beads.
11	Seeds and grammar	Build a bounded world from a recipe and novelty tokens.
12	Design	Child authors an accessible rule room and tests it with peers.

Facilitator rule

Never require all children to use the same channel, pace, role or product. Keep the learning question shared and the response routes flexible.

One lesson, multiple participation routes

Every lesson plan answers the same access questions before materials are distributed.

Inclusive lesson template

Planning field	Facilitator decision
Purpose	What substrate idea is being explored?
Meaningful choice	What real decision can every child make?
Perception routes	How can the objects and relationships be seen, heard, touched or summarized?
Action routes	How can choices be expressed without speech, reading, precision or two-hand movement?
Communication	Where are STOP, HELP, YES, NO, NOT SURE and CHANGE ROUTE?
Predictability	What sensory, timing, people and transition forecast is provided?
Repair	How can the child undo, correct an interpretation or leave?
Evidence	What child-selected artifact shows the idea without measuring disability?
Return	How will context be rebuilt after a break?

Plan for variation

Do not write a “normal lesson” and a separate special activity. Build several valid routes into the shared lesson from the start.

Family play without turning home into therapy

Definition Breaker can support shared curiosity and communication without asking family members to become clinicians or compliance trainers.

Follow interest

Use themes and objects the child already enjoys.

Keep sessions small

End while the child still has control; do not force completion.

Share roles

Adults also make mistakes, ask for help and use STOP.

Model language

Use WHERE, MATCH and READY naturally without quizzing every move.

Respect home communication

Use the child's established AAC, signs, objects and routines.

Celebrate decisions

Praise noticing, choosing, explaining, repairing and stopping - not speed or independence alone.

Protect privacy

Do not post access videos or communication moments without meaningful permission.

Bring questions back

Record barriers and preferences for designers or educators without blaming the child.

Family boundary

The game is play and learning. It does not replace communication assessment, occupational therapy, education plans or medical care.

Accessible assessment measures ideas, choices and explanations

Assessment accepts many forms of evidence and separates concept understanding from speed, speech, handwriting, gaze control or independent movement.

Equivalent evidence routes

Evidence option	What it may demonstrate
Place a tile or direct a partner	Identify support, compatibility or operation order.
Choose from AAC or object options	Classify a guard or repair.
Build a tactile replay chain	Understand event order and lineage.
Use screen-reader query	Inspect dependencies or first failed check.
Act out a topology route	Distinguish coordinate, sheet and orientation.
Design a room	Compose definitions, dependencies, guards and transitions.
Explain by speech, sign, drawing or symbols	Communicate reasoning through a preferred route.
Request STOP or CHANGE ROUTE	Exercise agency and metacognitive access knowledge.

Do not score access effort

Scan time, speech clarity, eye-gaze accuracy, motor precision, reading fluency and need for a partner are not part of the substrate-learning score unless they are the explicit and appropriate learning target.

Child co-design is an authority relationship

Children help define problems, routes, symbols, stories, priorities and release blockers. Participation itself is accessible and optional.

1 Invite

explain purpose, choices and privacy accessibly

2 Choose route

communication, sensory, motor and supporter preferences

3 Explore

let the child use and change the prototype

4 Decide

collect specific priorities and vetoes

5 Build

make visible changes and explain tradeoffs

6 Return

show what changed and ask whether it works

Real influence

At least some child decisions change the product, schedule or test plan.

Paid or recognized contribution

Where appropriate and lawful, value children's time without coercing participation.

Accessible feedback

Use objects, cards, AAC, observation, drawing, play and partner relay.

Right to disagree

Children can reject adult interpretations and decline future sessions.

Tokenism check

If children only choose colors after adults have fixed the rules, routes and release decision, the process is consultation at most - not co-design.

Safeguarding and assent in research and testing

Usability and co-design sessions require accessible information, meaningful assent, guardian processes where required, trained staff and a safe way to stop.

Safeguarding baseline

Requirement	Minimum practice
Accessible invitation	Explain purpose, activities, people, recording and choices in the child's route.
Assent	Look for active agreement and ongoing willingness; recheck when the activity changes.
Stop	Immediate, respected and available through the child's communication method.
No penalty	Payment, gift, school standing or game progress does not depend on finishing.
Data minimization	Collect only what is needed; separate access settings from research notes.
Recording	Use only with explicit bounded permission; provide nonrecorded participation where possible.
Supporter role	Support communication without answering for the child.
Distress plan	Pause, restore control, follow safeguarding protocol and do not resume by default.
Feedback return	Tell the child what was learned and what changed.

Professional boundary

This specification does not replace institutional ethics review, safeguarding law, school policy, clinical governance or qualified advice.

Representative child-access test scenarios

Scenarios combine game tasks with the child's communication, sensory, motor, cognitive and return needs.

Minimum representative scenarios

Scenario	Task	Evidence
A	Nonspeaking child uses AAC plus touch to reject a wrong interpretation.	NOT THAT cancels before commit; child selects replacement.
B	Blind child uses screen reader and one switch to solve Order Rail.	Graph queries, scan focus and replay remain synchronized.
C	Deafblind child uses object cues and tactile schedule, then requests a break.	Touch consent, orientation and return summary succeed.
D	Child with fluctuating fatigue changes from gaze to partner scan.	Candidate and focus persist; partner fidelity is confirmed.
E	Child with low vision and motor tremor uses enlarged select-then-place.	No drag required; accidental activation is repaired.
F	Child with cognitive support needs repeats Same Place room.	Same concept and reward remain; step list and examples are available.
G	Two children with different access routes share a commit.	Independent confirmations, STOP and equal credit.
H	Child returns after one week on a different device.	Context and access settings restore without a memory test.

Measure child agency and route quality

Metrics describe whether the system supports the child. They do not rank children or turn body data into ability scores.

Child-access metrics

Metric	Definition
Time to STOP	Time from need or prompt to successful pause through the child's route.
Meaningful choice rate	Confirmed child-directed choices divided by offered decision points.
Repair success	Misinterpretations or accidental actions resolved without lost progress.
Route-switch continuity	Percentage of switches preserving focus, candidate, context and reward.
Partner fidelity	Confirmed intents relayed accurately without option filtering.
Concept parity	Accessible route demonstrates the same rule relationships and outcome.
Effort ratio	Repetitions, navigation, time and force compared with the lowest-effort available route.
Return success	Child resumes meaningfully without a memory test.
Child comfort report	Child-selected or supporter-assisted report of control, clarity and willingness to continue.
Release blockers	Count and severity of essential actions lacking a complete access envelope.

No surveillance metric

Do not derive attention, emotion, compliance, intelligence or diagnosis from gaze, voice, movement, speed or help use.

A barrier log names the design, not the child

Barrier reporting should make the product fixable and avoid deficit descriptions of children.

Barrier-centered reporting

Avoid	Use instead
"Child failed the drag task."	"The room requires sustained drag; no select-then-place route exists."
"User was too slow."	"The scan timeout removed the option before the child confirmed."
"Child would not cooperate."	"The session did not provide an accepted STOP route and the child withdrew."
"Could not understand topology."	"The room represented sheets only by transparency and did not expose nonvisual relations."
"Needed too much help."	"The partner protocol was undefined, so adult actions were not traceable to child confirmation."
"Inaccurate eye control."	"Targets were smaller than the child's calibrated selection envelope and had no switch confirmation."
"Distracted by animation."	"Nonessential motion could not be reduced before entry."

Ownership

The team owns product barriers. The child owns their decisions, preferences and right to stop.

Release gates for child access

A release review checks the child experience, not only whether settings exist.

Minimum release gates

Gate	Pass evidence
Control	STOP and HELP work before calibration and from every route.
Perception	No essential relation is color-, sound-, motion-, transparency- or vision-only.
Operation	No essential action requires drag, hold, precision, simultaneous input or speech only.
Communication	YES, NO, NOT SURE, NOT THAT, WAIT and CHANGE ROUTE are respected.
Understanding	Purpose, consequence, first failed check and repair are available in plain and formal forms.
Agency	Partner assistance and adaptation remain traceable to child confirmation.
Continuity	Break, device loss and route switch preserve context and progress.
Parity	Accessible routes reach equal concepts, rewards, roles and endings.
Privacy	Access and communication data are minimal, private, editable and separate from lineage.
Physical safety	Prototype hazards and access force are reviewed before child use.
Validation	Children who use the routes participated in testing and fixes were retested.

Kill criteria: do not ship this version

The product is stopped, simplified or redesigned when any essential route violates child agency, safety or parity.

- STOP, HELP, privacy or access settings are unreachable before calibration, login or the first sensory event.
- An essential action has only one sensory channel or one motor gesture.
- Speech, eye contact, reading, handwriting, fast timing, two-hand movement or independent pointing is required without an equivalent route.
- A partner can choose for the child without explicit confirmation or can silently remove options.
- No-response, withdrawal, silence or compliance is treated as assent.
- Accessible play receives a lesser puzzle, reward, role, story or ending without an explicit player choice.
- Tracking loss, disconnect, break or route change loses focus, candidate state or progress.
- Forced touch, hand-over-hand movement, restraint or surprise sensory output appears in ordinary play.
- Raw gaze, voice, face, diagnosis, AAC content or access telemetry enters public logs, advertising or model training by default.
- A physical prototype lacks size, force, material, cord, magnet, battery, mouthing and damage review appropriate to intended users.
- The team treats adult simulation as a substitute for testing with children who use the access routes.
- The child cannot reject an interpretation, change route, undo or leave safely.

Authority

Any child tester, caregiver, accessibility specialist, safeguarding lead or product owner may raise a blocker. Severity is determined by harm and exclusion, not by how many users reported it.

Substrate source map

DB-1.0 combines suitable mechanisms without collapsing their evidence boundaries.

Substrate integration map

Source profile	Suitable contribution to Definition Breaker
UGTS-GN 1.1 GPU-native addendum	Canonical support -> compatibility -> guard -> event -> transition -> lineage sequence; typed state, precision, error and evidence boundaries.
UGTS-KC 2.0 Expanded Substrate	Versioned patterns, parametric paths, implicit fields, topology descriptors, kinematics, dynamics and hybrid-event resolution.
KC Two Hands 3.0	Stable scene identity, adapter-neutral hand state, two-hand transforms, proposals, conflicts, checkpoints and replay.
UGTS-KC 3.6	First-class definitions, content hashes, dependency closure, hinges and explicit phase or composition order.
SCLP 3.6.2	Guard intervals, phase and winding, reflective topology, constraint release, bounded grammar and finite spatial keys.
KC Elizabeth 3.9	Vector game scene, named input actions, deterministic fixed-step simulation, collision, animation, audio, persistence and offline HTML5.
SARA 3.6.3	Optional older-learner lesson on typed public/private data, permission and authorization; cryptographic and wallet operations remain outside ordinary play.

Evidence rule

Source-derived mechanics, engineering translations, accessibility requirements, educational scaffolds and optional research ideas remain visibly distinct.

External accessibility guidance crosswalk

DB-1.0 aligns its testing language with widely used accessibility, learning, gaming, communication and child-rights guidance while keeping product-specific evidence requirements.

Guidance crosswalk

Guidance family	Applied design themes
W3C WCAG 2.2	Perceivable alternatives, keyboard access, focus, target size, dragging alternatives, motion and flash safety, status messages and accessible authentication.
W3C cognitive-accessibility guidance	Clear purpose, familiar language, help, error prevention, personalization and reduced memory burden.
CAST UDL Guidelines 3.0	Multiple means of engagement, representation, action and expression; learner agency and identity.
Xbox Accessibility Guidelines	Input remapping, text and subtitles, narration, visual and audio cues, motion, difficulty and accessible testing.
AAC professional guidance	AAC supplements or replaces speech; communication systems include symbols, devices and partner practices.
Deafblind education guidance	Do with, not for; hand-under-hand; anticipation, object cues and adequate pace, process and wait time.
UN child and disability rights	Participation, dignity, privacy, accessible information and the right to play.
UNICEF inclusive digital learning	Accessible formats and learning materials designed for children with and without disabilities.

No conformance claim

Using these sources does not certify the future game, PDF, toy or curriculum. Conformance requires product-specific testing, evidence and qualified review.

Production roadmap

The work is staged so that access evidence appears before content scale and physical manufacturing investment.

Production roadmap

Phase	Deliverable	Exit condition
0 - Co-design setup	Advisory group, safeguarding, communication routes, Access Passports.	Children can participate and stop safely.
1 - Low-tech prototype	Rule tiles, order rail, try/real mats, STOP board and replay chain.	Core loop works through multiple physical routes.
2 - Digital semantic slice	One room with complete object semantics and named actions.	Keyboard, screen reader, touch and switch pass.
3 - Combined access	Gaze, AAC, partner scan, quiet mode and route switching.	Representative combined scenarios pass.
4 - Learning bridge	Explorer, Builder and Designer explanations plus portfolio.	Concept parity is demonstrated across routes.
5 - Game vertical slice	Three rooms, narrative, audio, saves and offline build.	Release gates pass with child testers.
6 - Physical product research	Material, force, size, mounting and hazard prototypes.	Qualified safety and accessibility review supports next step.
7 - Scale and workshop	Authoring lint, localization, community rooms and moderation.	Accessible creation and sharing are validated.

Document accessibility statement

The document is generated from structured HTML with semantic headings, real text, table headers, internal links, bookmarks and language metadata. It targets a tagged PDF/UA-1 output profile.

Readable design High contrast, left-aligned text, generous spacing, consistent page structure and no information carried by color alone.	Navigation Bookmarks and linked contents support rapid movement through the document.
Searchable text Text remains selectable and searchable; diagrams are built from tagged text and CSS rather than flattened screenshots.	Printable tools Prototype boards use large words, strong borders and redundant patterns.
Known limitation Tagged output has been technically inspected but not independently certified against PDF/UA or every assistive-technology combination.	Product boundary The PDF cannot itself validate child development, curriculum outcomes, clinical methods, toy safety or regulatory compliance.
Preferred source For future revisions, preserve the structured HTML source and re-run tagged PDF generation rather than editing page graphics or flattening text.	

Final formal decision

Definition Breaker DB-1.0 is accepted as the child-access standard and inclusive play system for the concept. The authoritative learning loop remains the real UGTS sequence, while every child may reach it through a route that fits their perception, movement, communication, understanding, regulation and energy.

Same rules
All routes enter one semantic proposal and verification pipeline.

Same agency
Children can stop, refuse, ask, confirm, repair, change route and return.

Same depth
Child language grows into formal definitions, dependencies, topology, event calculus and lineage.

Same dignity
No baby mode, public deficit label, lower reward or hidden adult control.

Physical continuity
Toys and tabletop games use the same IDs, relations, try/check/commit states and replay ideas.

Evidence before claims
Accessibility, learning, performance and safety claims require measured, participatory validation.

WHERE? -> MATCH? -> READY? -> CHECKED? -> CHANGE -> REMEMBER
Every child can enter, understand, act, repair and grow within the same rule world.

Status
Ready for participatory prototype development. Not yet a certified educational product, clinical intervention, PDF/UA-conformant publication or regulated toy.

Core communication board

Point, look, touch, scan, sign, speak or ask a partner. Keep this board available throughout play.

STOP
Pause now. Keep my place.

HELP
Explain or offer another route.

WAIT
Give me more time.

NOT THAT
That was not my choice.

AGAIN
Show, say or present it again.

DIFFERENT
Change route or offer another option.

BREAK
Save my place and pause.

DONE
Save and finish safely.

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Yes, no, not sure and stop cards

Cut or mount as large cards. Add the child’s tactile marks, symbols, language or AAC references.

YES

That is my choice.

NO

Do not do that.

NOT SURE

Keep it in preview or tell me more.

STOP

Pause now.

WAIT

Give me time.

CHANGED MIND

Undo and let me choose again.

My reliable YES signal is:

My reliable NO signal is:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Choose my access route

Choose any combination. Circle, point, place a marker, gaze or ask a partner to relay.

LOOK

Large, clear, high-contrast information.

LISTEN

Speech, audio summary or earcons.

TOUCH

Objects, texture, braille or raised lines.

POINT / GAZE

Finger, eye, head or pointer.

PRESS / SCAN

Key, switch, dwell or scanning.

SIGN / AAC

Sign, symbol, device, book or custom signal.

PARTNER

Names or moves only my confirmed choice.

QUIET / SLOW

Less sound, motion, choices or pressure.

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

First, then, next, break, finished

Place an object, symbol or word in each box. Show the whole plan while presenting one step at a time.

FIRST

What I do now

THEN

What comes after

NEXT

•The following step•

BREAK

Pause and keep my place

FINISHED

Save and close

CHANGE

A different route or plan

What will stay the same?

What will change?

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

One-switch scan board

A simple one-switch scan order. The facilitator or software highlights one item at a time and waits for the child’s agreed signal.

1 - NEXT

Move to the next object or choice.

2 - CHOOSE

Select the current object.

3 - PREVIEW

Tell me what would happen.

4 - CHECK

Ask the rule questions.

5 - UNDO

Return to the last clear state.

6 - STOP

Pause and keep my place.

7 - HELP

Explain or change route.

8 - REST

Wait here without choosing.

Scan speed or wait time:

Confirmation method:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

The six rule-question cards

These are the six child-language questions for the complete substrate. Cut as cards or keep as one board.

WHERE?

Can this rule reach the object or place?

MATCH?

Are the types, sheets, roles and policies compatible?

READY?

Has the boundary or confirmation condition been met?

CHECKED?

Did the whole proposal pass its checks?

CHANGE

What exact part of the world changes?

REMEMBER

What must be replayed or restored?

WHERE -> MATCH -> READY -> CHECKED -> CHANGE -> REMEMBER

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Blank rule tiles

Make custom rule objects. Keep the same stable ID across physical and digital versions.

RULE NAME

ID: _____
TYPE: _____
NEEDS: _____

RULE NAME

ID: _____
TYPE: _____
NEEDS: _____

RULE NAME

ID: _____
TYPE: _____
NEEDS: _____

RULE NAME

ID: _____
TYPE: _____
NEEDS: _____

What must stay true?

How will a child identify this without color?

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Tactile symbol planning sheet

Plan a redundant symbol system. Test each distinction with children who will use it.

Tactile and multisensory symbol plan

Rule class	Word	Shape	Border	Texture / raised mark	Spoken or haptic cue
Support	WHERE	_____	_____	_____	_____
Compatibility	MATCH	_____	_____	_____	_____
Guard	READY	_____	_____	_____	_____
Verified event	CHECKED	_____	_____	_____	_____
Transition	CHANGE	_____	_____	_____	_____
Lineage	REMEMBER	_____	_____	_____	_____
Uncertain	UNCLEAR	_____	_____	_____	_____

Child feedback and changes:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

The diagram illustrates a four-stage process for building a model, arranged in a 2x2 grid. Each stage is represented by a rounded rectangle with a unique background pattern and a set of instructions.

- TRY** (Top Left): Background is a grid of small dots. Text: "Build or place the candidate here. Nothing is real yet."
- CHECK** (Top Right): Background is diagonal lines. Text: "WHERE? MATCH? READY? Any uncertainty or conflict?"
- REAL** (Bottom Left): Background is solid white. Text: "Move here only after the child confirms the checked change."
- REMEMBER** (Bottom Right): Background is a grid of larger squares. Text: "Add the event card, bead or replay record."

At the bottom of the diagram, a yellow bar contains the sequence: **TRY -> CHECK -> REAL -> REMEMBER**.

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Partner-assisted scanning record

Use this sheet to keep partner-assisted scanning stable and child-directed.

Partner scan protocol

Protocol field	Agreed method
Child YES signal	_____
Child NO signal	_____
Child STOP signal	_____
Child WAIT / more-time signal	_____
Choice presentation order	_____
Typical wait time	_____
Confirmation wording	_____
How to show uncertainty	_____
What the partner must never do	_____
How to repair a misunderstanding	_____

Child approved this protocol by:

Review date or trigger:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Sensory and effort forecast

Show this before a room, lesson, toy or group activity. Circle or mark the child’s choices.

Sensory and effort forecast

Channel	Forecast	Child choice
Sound	quiet / music / voice / effects / sudden changes	off / quiet / captions / tactile / other
Light	bright / dim / glow / contrast change	soft / high contrast / no glow / other
Motion	still / animation / camera / scrolling	still / reduced / slow / other
Touch	objects / vibration / partner touch	off / gentle / object / ask first / other
Time	untimed / scan / countdown / repeated prompts	slow / pause / practice / other
People	solo / partner / group / turn-taking	observe / quiet role / co-play / other
Effort	choices, reach, force, repetitions and duration	shorter / mounted / partner / other

STOP and break route:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Return after a break

A return page prevents breaks from becoming memory tests.

WHERE I WAS

Room, goal and current object

WHAT CHANGED

Important committed events

WHAT I WAS TRYING

Candidate proposal

WHAT CAN HAPPEN NEXT

Continue, undo, review, change route or finish

My preferred return summary:

Access routes to restore:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Child co-design feedback board

Children can use words, symbols, pointing, AAC, objects, switches or partner relay to complete this board.

KEEP What should stay the same?	CHANGE What should be different?
STOP What should not happen again?	TRY NEXT What shall we test?
MORE What needs more detail or challenge?	LESS What needs less sound, motion, effort or information?

My most important decision:

How the team will show what changed:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Body and energy check-in

Use this as a pre-session and post-session check. The child may point, gaze, choose an object, sign, speak, scan or ask a partner to relay.

READY I can begin or continue.	SLOW Use fewer steps and more time.	QUIET Reduce sound, motion and choices.
SUPPORTED Use positioning, switch, gaze or partner help.	BREAK Save my place now.	DONE Finish safely.

Anything else my body or energy needs today:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

My Access Passport - printable

A privacy-safe, child-owned record of preferred access. Complete only what is useful.

My Access Passport

Area	My choice
How I say YES / NO / STOP	_____
How long I may need	_____
How I receive information	_____
How I choose and move	_____
Touch and positioning	_____
Sound, light, motion and vibration	_____
What helps when tired or overloaded	_____
Partner confirmation rules	_____
What must stay private	_____
How I change or delete this passport	_____

I reviewed this on:

My chosen review or change signal:

Prototype access sheet. Adapt language, symbols, tactile marks, mounting, size and spacing with the child. Keep a low-tech copy available. Do not laminate until the child has helped test the arrangement.

Evidence classes and claim discipline

The document separates what the supplied substrate reports establish from what DB-1.0 proposes for game design, accessibility, education and physical product development.

Evidence-class ledger

Class	Contains	How it is used
Source-derived	Mechanics and boundaries explicitly present in the supplied UGTS reports.	Retained with version attribution and original scope.
Engineering translation	Data models, adapters, state machines, interfaces and measurable requirements that preserve the source meaning.	Specified for implementation and validation.
Accessibility design requirement	Child agency, communication, sensory, motor, cognitive, privacy and parity rules.	Required for Definition Breaker, informed by guidance and co-design.
Educational scaffold	Child language, lesson arcs, toys, printables, roles and assessment routes.	Prototype and validate with children, families and educators.
Optional research idea	New interactions or physical products that may be valuable but lack product evidence.	Label clearly; do not present as complete or certified.
Rejected escalation	Magic compression, one-bit total state, topology-created energy, forced touch, inferred consent or diagnosis.	Excluded from requirements and claims.

Trust rule

A persuasive metaphor never becomes a mathematical, physical, educational, clinical or accessibility fact without an explicit model and suitable evidence.

Core glossary: access and agency

These terms are used consistently throughout DB-1.0.

Access glossary

Term	Definition
Access route	A working combination of perception, understanding, action, communication, confirmation, repair and return.
Access envelope	The six-stage test: perceive, understand, act, confirm, repair and resume.
AAC	Augmentative and alternative communication that supplements or replaces speech through symbols, signs, devices, objects and partner practices.
Assent	The child's accessible, ongoing willingness to participate; distinct from silence, compliance or guardian permission.
Child-directed	The child controls meaningful choices, stop, route, confirmation and repair, including when support is used.
Concept parity	Different access routes teach and evaluate the same relationships and rules.
Dignity parity	Accessible routes avoid lower status, childishness, public labels, pity and unnecessary adult control.
Partner fidelity	A supporter presents options and relays only confirmed child intent without filtering or substitution.
Route-switch continuity	Focus, candidate, context and reward survive a change of device, communication or sensory route.
Supported decision-making	Help that improves access to information and expression while keeping the choice with the child.

Core glossary: substrate

Child language and formal language refer to one shared architecture.

Substrate glossary

Child term	Formal term	Meaning
WHERE?	Support	Whether a rule or object is locally relevant to the query.
MATCH?	Compatibility	Whether types, sheets, phases, roles and policies may couple.
READY?	Guard	Whether a threshold, boundary, timing or confirmation condition is accepted.
CHECKED?	Verified event	A complete proposal that passes declared checks and conflict policy.
CHANGE	Transition	An atomic state patch applied after verification.
REMEMBER	Lineage	Ordered event history, identity, checkpoints and replay.
NEEDS	Dependency	A definition or object required before another can be evaluated.
RULE CARD	Definition node	A typed, versioned, content-addressed rule record.
ORDER RAIL	Evaluation schedule	The explicit order of parsing, transforms, checks, commit and lineage.
SHEET	Topology state	A layer or quotient class that distinguishes co-located states.
TRY WORLD	Candidate state	A reversible proposal separate from authoritative state.
REPLAY BEADS	Event log	A physical representation of ordered committed transitions.

Supplied substrate references

These project documents provide the formal and implementation basis used by DB-1.0.

Project source register

Document	Role in DB-1.0
Unified Geometric-Topological Substrate GPU-Native Addendum, UGTS-GN 1.1	Canonical query-first sequence, typed state, guard and event logic, memory and evidence boundaries.
UGTS-KC 2.0 Expanded Substrate	Pattern families, parametric and implicit geometry, topology, kinematics, dynamics and hybrid events.
UGTS-KC Two Hands 3.0 Interactive Graphics Runtime	Scene identity, interaction adapters, two-hand calculus, proposals, commit, conflict, checkpoints and replay.
UGTS-KC 3.6 Literal Referential Substrate	Definitions as data, dependency closure, content hashes, hinges, profiles and explicit operation order.
UGTS-KC 3.6.2 SCLP	Packed finite state, guard intervals, winding, reflective topology, constraints, grammar and 64-bit keys.
UGTS-KC 3.6.3 SARA	Optional public/private and authorization lesson; strong secret and wallet-operation boundary.
UGTS-KC 3.9 KC Elizabeth Vector Game Runtime	Vector scene, input, collision, animation, audio, fixed-step simulation, persistence and offline browser delivery.
Definition Breaker DB-0.1 through DB-0.6	Concept, all-ages learning, physical-play, child-first, universal and multisensory accessibility lineage.

Attribution boundary

Requester attribution appearing in source reports is not independently verified by this document. DB-1.0 uses the technical content and supplied filenames as project references.

External guidance references

The following official guidance families informed the accessibility and participation requirements. Their use does not constitute endorsement or certification.

External guidance register

Organization or instrument	Reference used
World Wide Web Consortium	Web Content Accessibility Guidelines (WCAG) 2.2 and cognitive-accessibility guidance.
CAST	Universal Design for Learning Guidelines 3.0.
Microsoft Gaming Accessibility	Xbox Accessibility Guidelines, including input, text, audio, motion and testing topics.
American Speech-Language-Hearing Association	Augmentative and Alternative Communication practice guidance.
National Center on Deafblindness	Communication, hand-under-hand, object cues, anticipation and wait-time learning resources.
United Nations	Convention on the Rights of the Child and Convention on the Rights of Persons with Disabilities.
UNICEF	Digital inclusion and accessible learning-material guidance for children with disabilities.
PDF Association / ISO ecosystem	PDF/UA principles for tagged, navigable and assistive-technology-compatible PDF documents.

Current-state note

Guidance and standards may be revised. Product teams recheck official versions during implementation and before release.

Revision lineage

Definition Breaker accessibility evolved through repeated child-access expansions. DB-1.0 consolidates that work into a coherent standard and play system.

Revision history

Version	Primary contribution
DB-0.1	Initial game concept: definitions and rules as physical world objects.
DB-0.2	All-ages learning path and physical-play scaffolding.
DB-0.3	Child-first easy-read route, access passport, printable access concepts and tagged-PDF target.
DB-0.4	Child-led universal access, AAC, nonvisual architecture, partner-assisted scanning and fairness rules.
DB-0.5	Child-ready first game, printable toolkit, broader communication routes and agency requirements.
DB-0.6	Multisensory and combined access, deafblind and profound-access pathways, physical-access variants and child-defined metrics.
DB-1.0	Access envelope, full route parity, expanded communication autonomy, complete substrate learning bridge, digital product standard, physical toy system, co-design evidence and release blockers.

Supersession

DB-1.0 supersedes earlier accessibility editions while retaining every prior requirement that remains more protective or child-directed.

Implementation handoff checklist

A concise checklist for the first production sprint.

First-sprint handoff

Owner	First implementation commitment
Product	Adopt access envelope, parity and kill criteria as release requirements.
Game design	Build Order Rail and Same Place rooms with candidate/check/commit/replay.
Engineering	Implement semantic object contract, named actions, AccessProfile and AccessSessionState.
Accessibility	Recruit paid or appropriately recognized child advisors using relevant access routes.
UX writing	Create child, facilitator and formal explanations plus constructive reason codes.
Audio	Caption and transcribe; build optional stable earcon legend and quiet defaults.
QA	Automate lint and run combined-access scenarios with route-switch continuity.
Privacy	Separate access data from world lineage; prohibit raw gaze, speech and diagnosis inference.
Education	Prototype the twelve-session arc and equivalent assessment routes.
Physical design	Build low-cost tiles, order rail, try/real mat and replay chain; begin force and hazard review.
Safeguarding	Approve assent, stop, recording, supporter and distress protocols.
Leadership	Block content scale until the representative vertical slice passes child-access gates.